

UntrashOS on the Zyxel NAS542 — User Guide

Installation and operation of UntrashOS on the Zyxel NAS542

This guide covers installing UntrashOS from the USB stick and every function of the **Untrash Desk**, the web interface you use to run the NAS. Each screen is described with all of its options. The tables list each option, its choices (the default is **bold**), and what it does.

> **About the screenshots.** The screenshots were taken on *emulated* hardware: the installer runs in its dry run (commands are shown, not executed), and the Desk runs in a test mode with example data. All names, addresses and files are examples (users `anna` and `ben`, network `192.0.2.x`, `*.example.net`).
>
> The guide is also built into UntrashOS: open **Manual** in the Desk, use the link on the sign-in page, or the link at the top of the installer. All of them offer this guide as a PDF as well.

Contents

1. Installation
2. First start and sign-in
3. Working with the Desk
4. Files
5. Shares
6. Storage
7. Users and groups
8. Network
9. Settings
10. Ad filter
11. Media server and channels
12. Cloud
13. Downloads
14. Backup
15. Updates
16. Status, logs and system tools
17. Working as a normal user
18. Back to the Zyxel firmware

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1. Installation

1.1 What you need

- A Zyxel NAS542 running the original Zyxel firmware, with its disks in place.
- A USB stick of at least 4 GB (the image is 3.4 GB). **It is erased.**
- A PC, tablet or phone with a web browser on the same network as the NAS.
- The UntrashOS stick image `untrash-stick-<version>.img`.

1.2 Write the stick

Write the image to the stick with any image writer (balenaEtcher, Raspberry Pi Imager, `dd`). On Linux, the project script does the same with extra checks. Replace `/dev/sdX` with your stick. Every byte on that device is overwritten:

```
sudo install/write-stick.sh untrash-stick-<version>.img /dev/sdX
```

The stick image is the same for every NAS542. Nothing on it is made for a particular device. Everything that belongs to one NAS is created on that NAS during the first run. The stick then holds that NAS's backup and is marked as used. **Use a freshly written stick for each NAS.**

1.3 First run under the Zyxel firmware

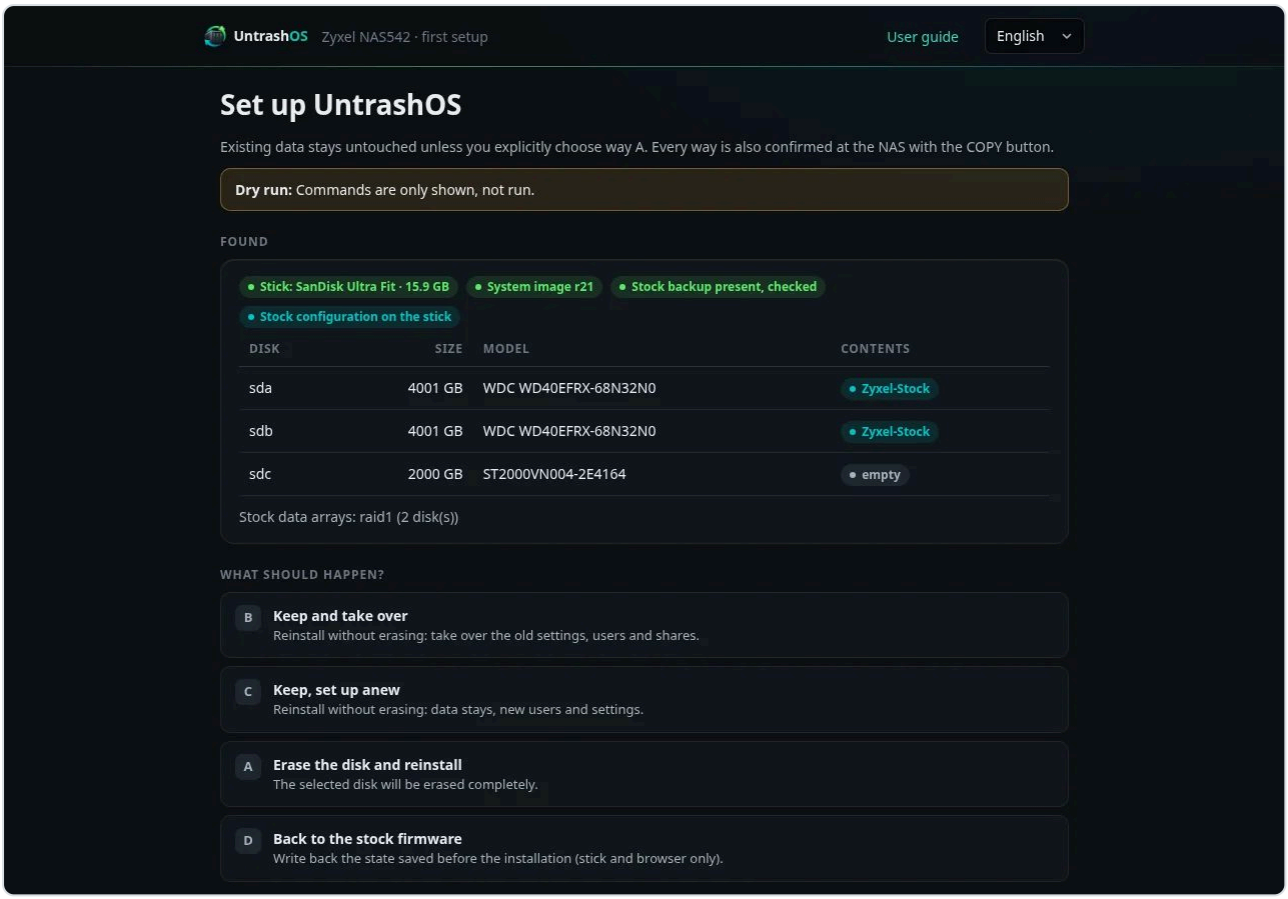
1. Switch the NAS off and plug the stick into a USB port of the NAS.
2. Switch the NAS on. While it starts, the Zyxel firmware runs the install script from the stick:
    1. It checks that the flash layout and the boot settings are those of a normal NAS542.
    2. It saves a **complete backup of the flash** to the stick, reads it back and compares it.
    3. It saves the **Zyxel configuration** (users, groups, shares, passwords) to the stick.
    4. It installs the small UntrashOS loader into the unused one of the two firmware slots and adds one call to the boot settings. The Zyxel firmware itself is not changed.
    5. It **switches the NAS off**.
3. Leave the stick in the NAS.

If any step fails, the script writes the backup back, leaves an `INSTALL_FAILED` marker on the stick and switches the NAS off. The NAS then starts the Zyxel firmware as before.

1.4 Open the installer

Switch the NAS on again. The loader starts the installer from the stick. After a minute or two, open

http://untrash.local/ in the browser. If your router does not resolve `.local` names, try `http://untrash/` or the address your router lists for the NAS. If another UntrashOS device already uses the name, the installer answers as `http://untrash-2.local/` (then `-3`, ...).



Element	Meaning
Stick	Model and size of the installation stick.
System image	The UntrashOS version on the stick. *Missing* means the stick image is incomplete.
Stock backup	*present, checked*: the flash backup was read back and compared when it was made. *present (unchecked)*: the backup exists but was not compared. *No stock backup*: way D is not possible.
Stock configuration on the stick	The saved Zyxel users, groups and shares, needed for way B.
Disk table	Every disk with size, model and content: **Zyxel-Stock** (a disk of the Zyxel firmware), **empty**, **UntrashOS**, **RAID member** or **unknown**.
Language (top right)	English, Deutsch, Français. The installed system starts with the language chosen here.
User guide (top)	Opens this guide.

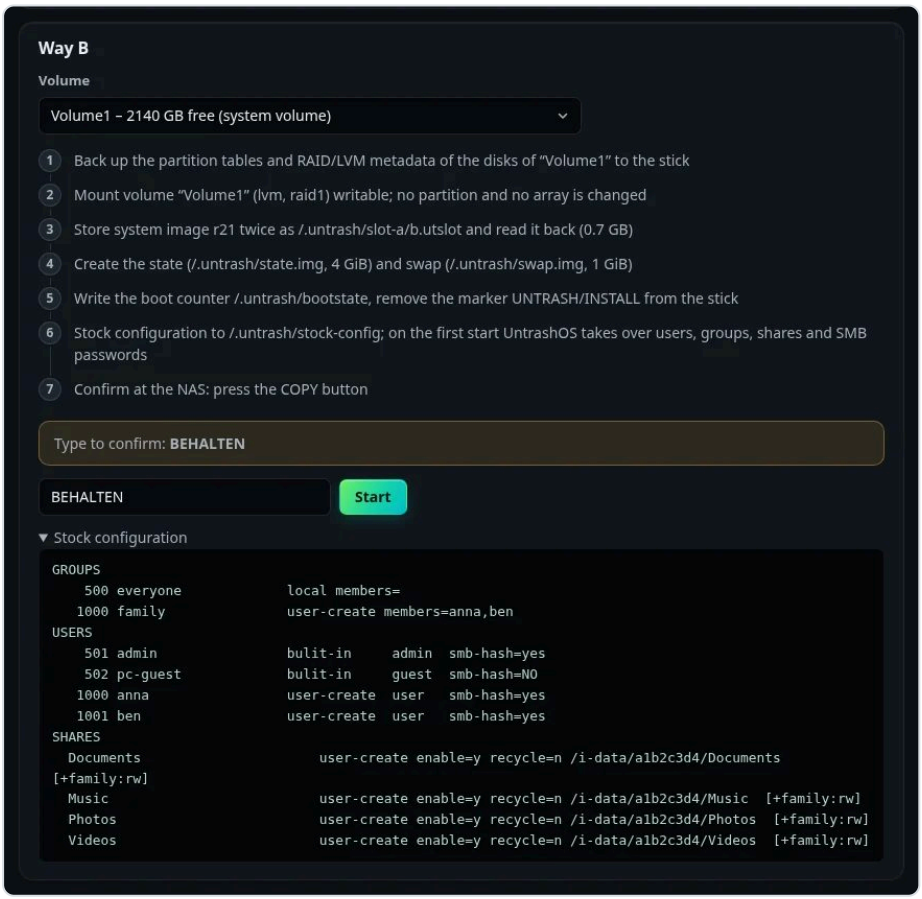
1.5 The four ways

There are four ways. **Ways B and C keep all your data**. They change no partition and no RAID array. UntrashOS is stored as a few files in the hidden folder `/.untrash` of an existing Zyxel volume. Only **way A** erases a disk.

Way	What happens	Data	Confirmation word
B – Keep and take over	New installation on a Zyxel volume. Users, groups, shares and SMB passwords of the Zyxel firmware are taken over on the first start.	kept	`KEEP`
C – Keep, set up anew	New installation on a Zyxel volume. A new `admin` password and a network name are set. Every top-level folder becomes a share for `admin`. The old users are not taken over.	kept	`KEEP`
A – Erase the disk and reinstall	The chosen disk is erased and gets its own UntrashOS layout (two system slots, state, swap, data area with btrfs).	**erased** on that disk	`DELETE <disk>`, e.g. `DELETE sdc`
D – Back to the stock firmware	Writes the Zyxel boot data back from the backup on the stick (see section 18).	kept	`STOCK`

Choosing a way shows its **plan**, step by step. Nothing happens until you confirm:

1. Type the confirmation word shown in the yellow box and click **Start**.
2. **Press the COPY button on the front of the NAS within 60 seconds.** Without the button press, the installer stops and changes nothing.

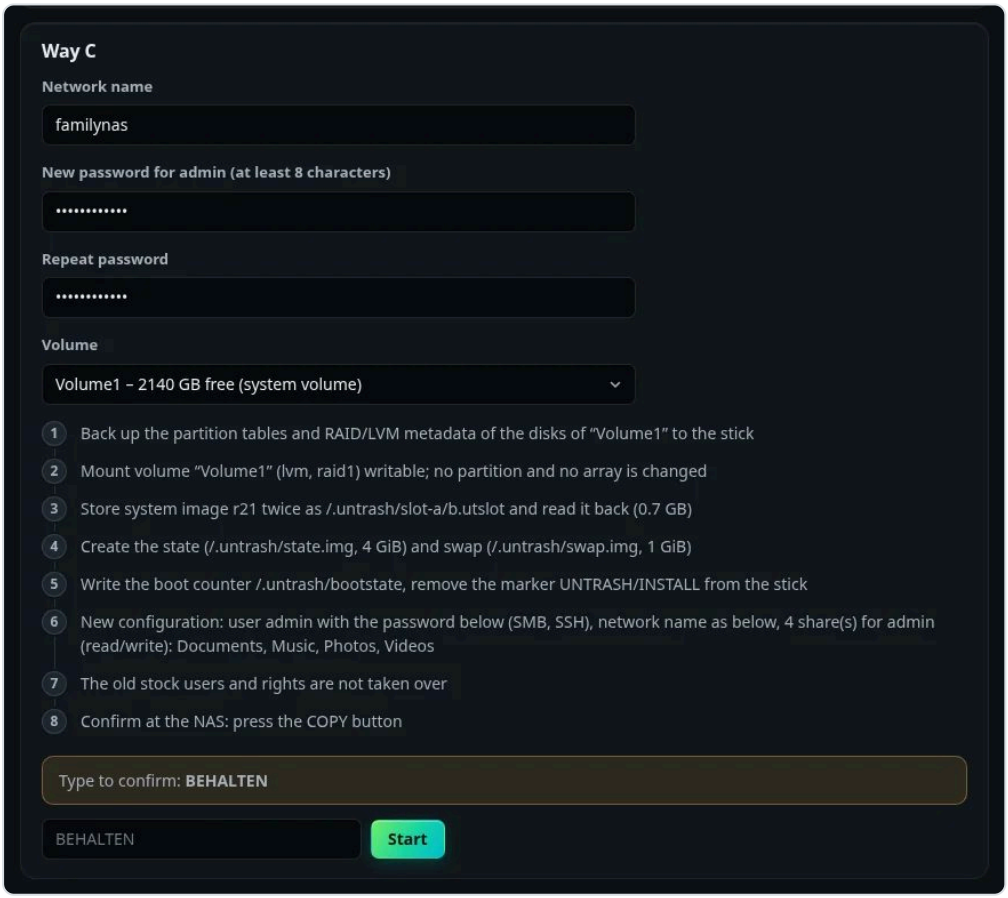


Way B options

Option	Choices	Meaning
Volume	the Zyxel volumes, **system volume first**	Where the UntrashOS files go. The installer shows the free space. It needs room for two copies of the system image plus 5 GiB.
Stock configuration	(expandable)	Shows the groups, users and shares that will be taken over.

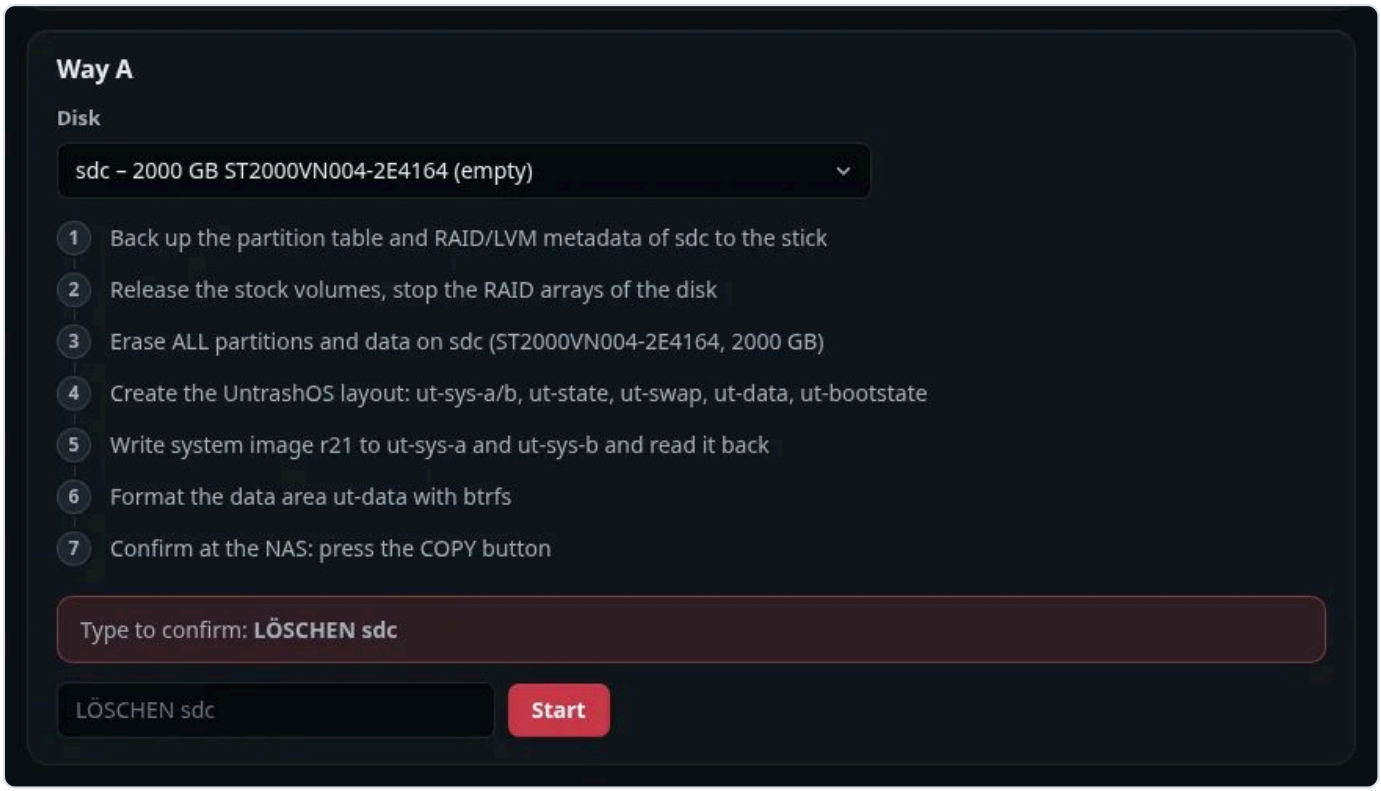
Way C options

Option	Choices	Meaning
Volume	as for way B	Where the UntrashOS files go.
Network name	text, default **untrash**	1–15 letters, digits or hyphens. The NAS is reachable as `<name>.local`.
New password for admin	text	At least 8 characters. Used for the Desk, SMB and SSH.
Repeat password	text	Must match.



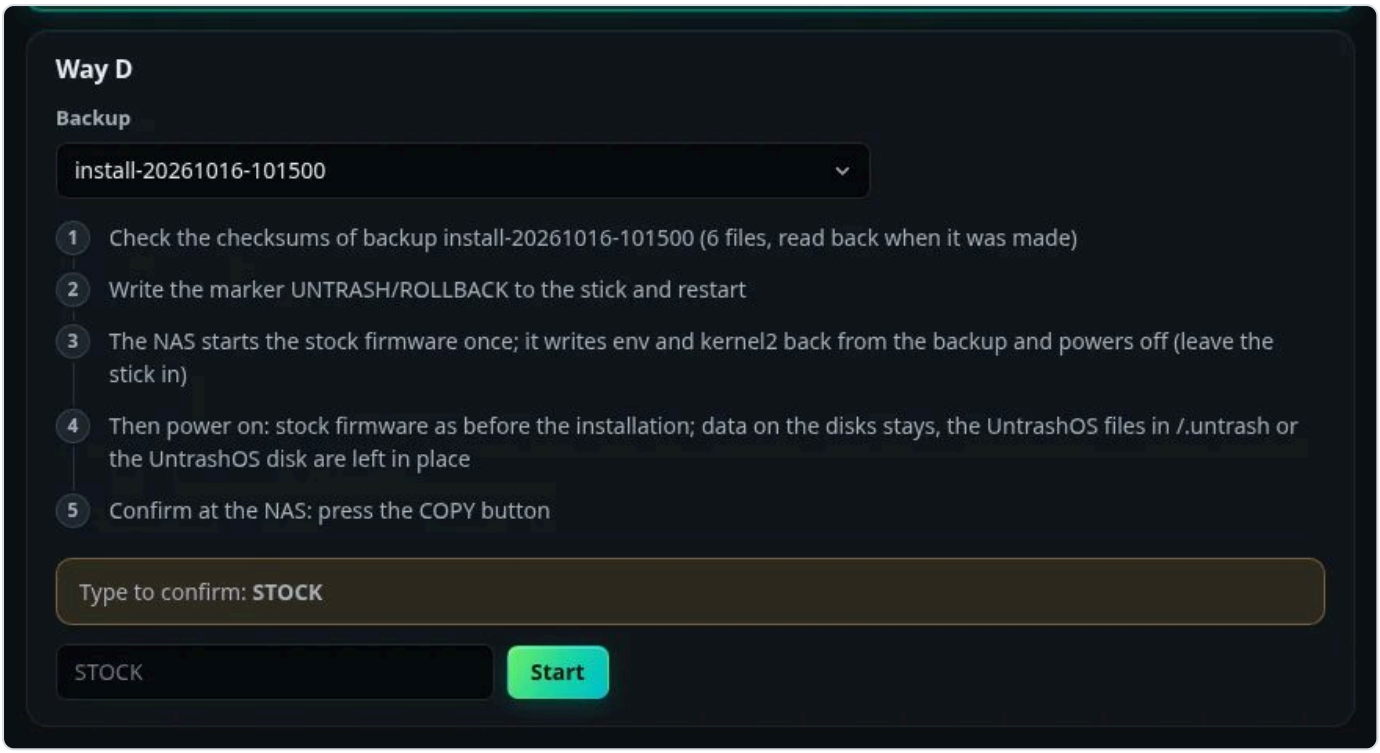
Way A options

Option	Choices	Meaning
Disk	every disk with size, model and content	The disk to erase. It needs at least 16 GiB. The disk the installer itself runs from cannot be chosen. If the disk belongs to a Zyxel RAID array, the plan warns that the array loses a disk.



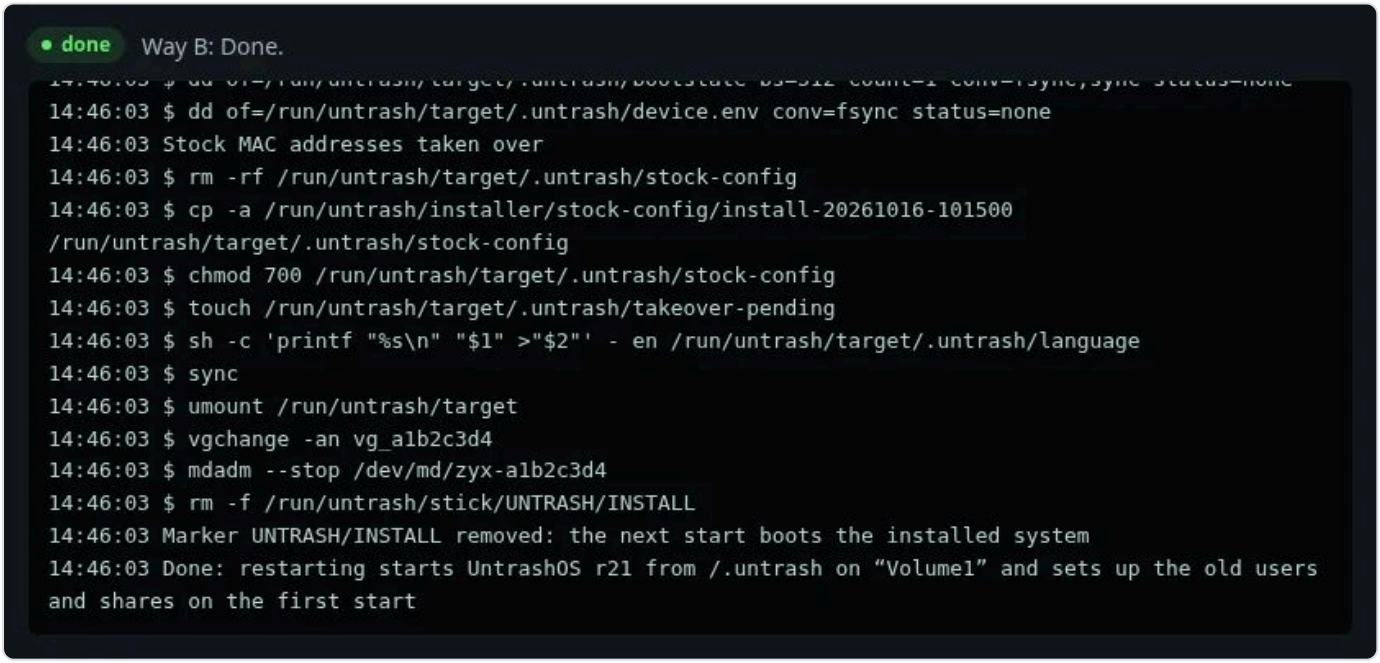
Way D options

Option	Choices	Meaning
Backup	the completed installation backups on the stick, **newest first**	The state the NAS returns to.



The installer refuses to start when a requirement is missing, and says why. Examples: no stick or no system image, too little free space on the volume, an incomplete RAID array ("insert all disks first"), a disk that already carries UntrashOS partitions while you choose way B or C.

1.6 Progress and restart



The **Progress** box shows the state (*ready*, *COPY button*, *running*, *done*, *error*) and logs every step and command. Before anything is written, the partition tables and RAID/LVM metadata of the affected disks are saved to the stick (`UNTRASH/disk-backup-…`). Every system image that is written is read back and compared.

When the state is **done**, click **Restart**. The NAS starts the installed UntrashOS. You can remove the stick afterwards. **Keep it:** it holds the backup for the way back to the Zyxel firmware.

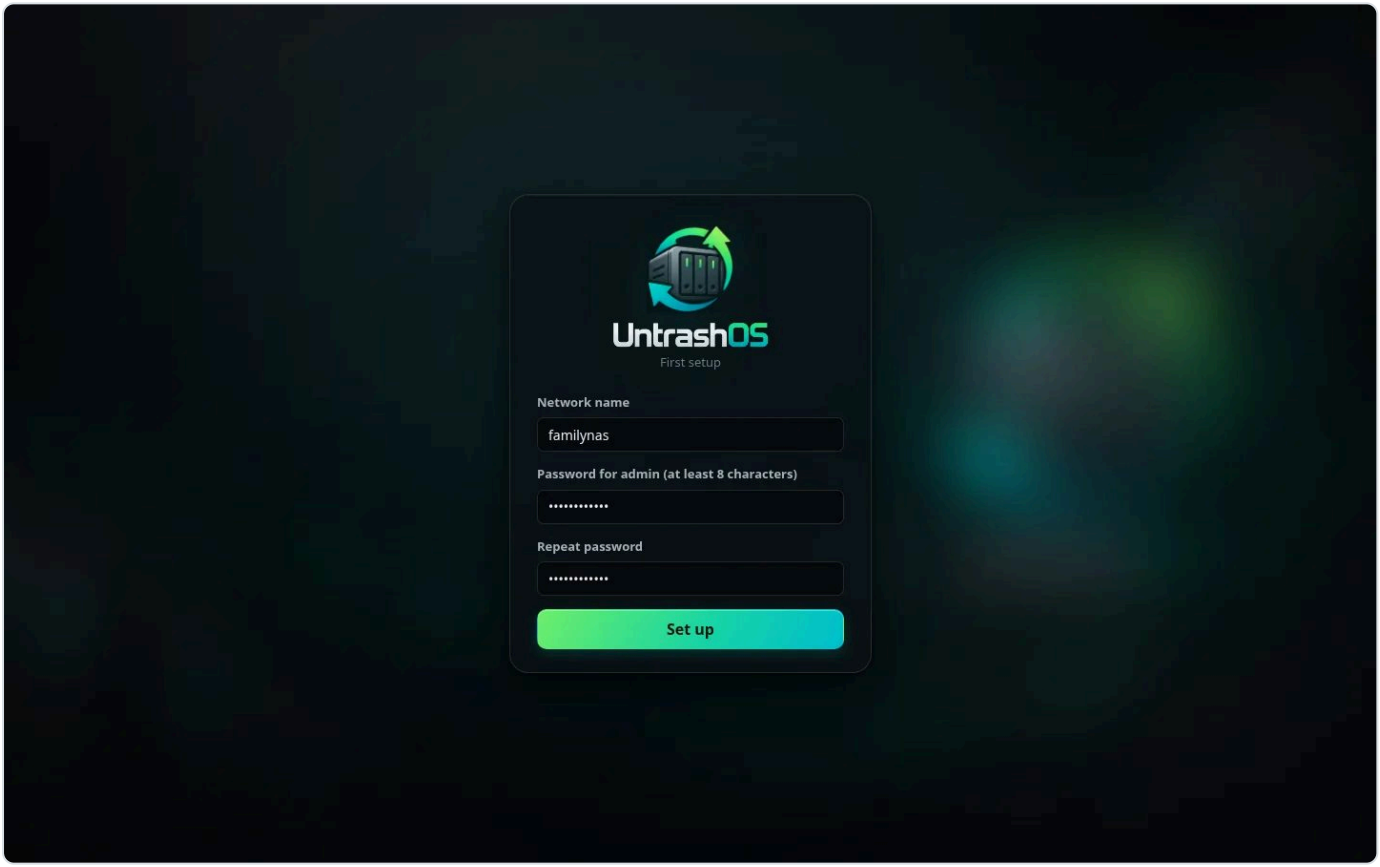
> The yellow *Dry run* box and `(DRY RUN)` in the screenshot exist only in the emulation used for this guide.

2. First start and sign-in

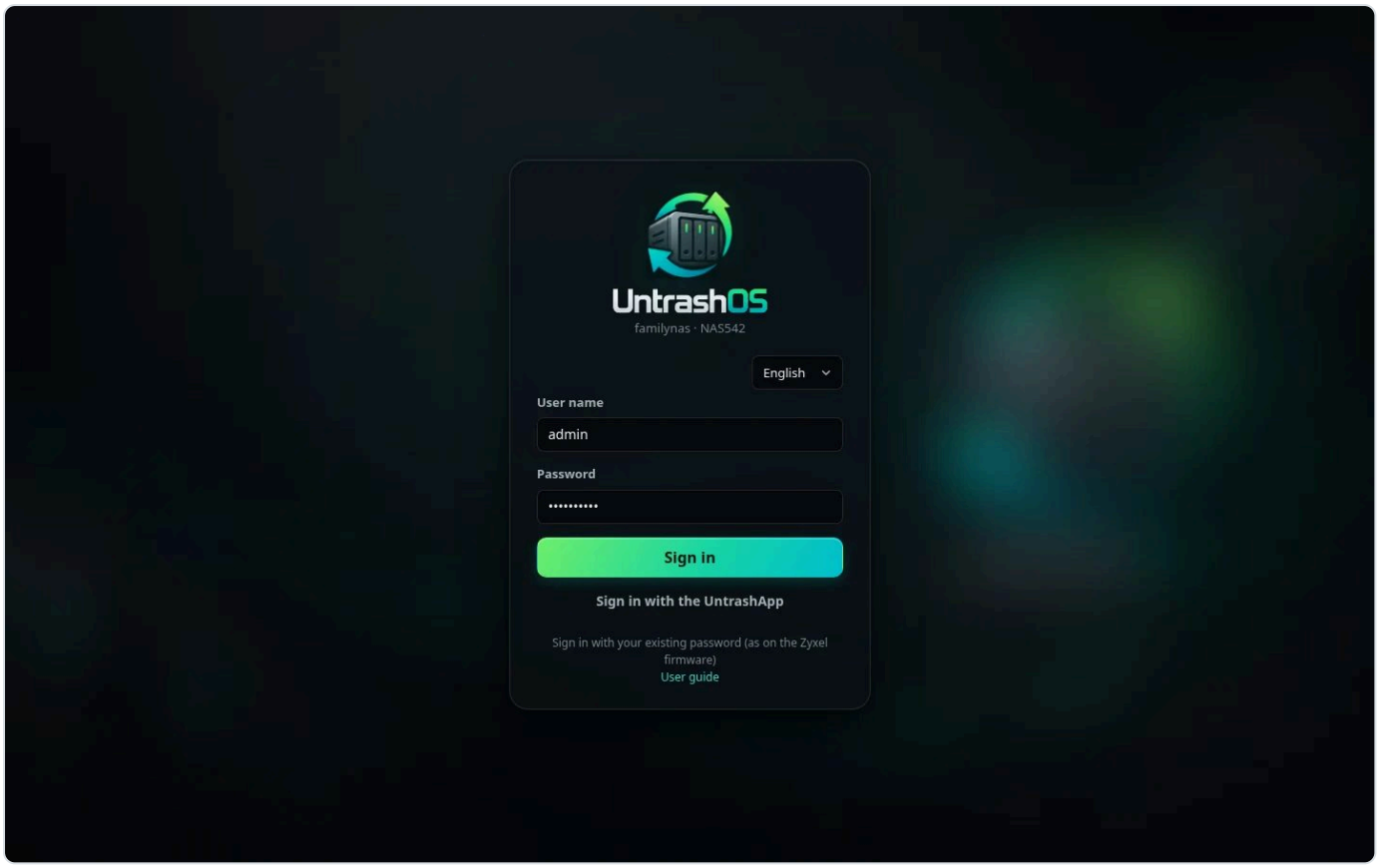
Open `http://<network name>.local/` (for example `http://untrash.local/`), or the address of the NAS. A NAS with the default name that finds `untrash` taken in the network becomes `untrash-2` (then `untrash-3`, …) and keeps that name from then on; a name you chose, or the name of the stock system (way B), is never changed.

After way A the NAS has no users yet. The first page asks for:

Option	Meaning
Network name	Host name of the NAS (1–15 letters, digits, hyphens).
Password for admin	At least 8 characters.
Repeat password	Must match.

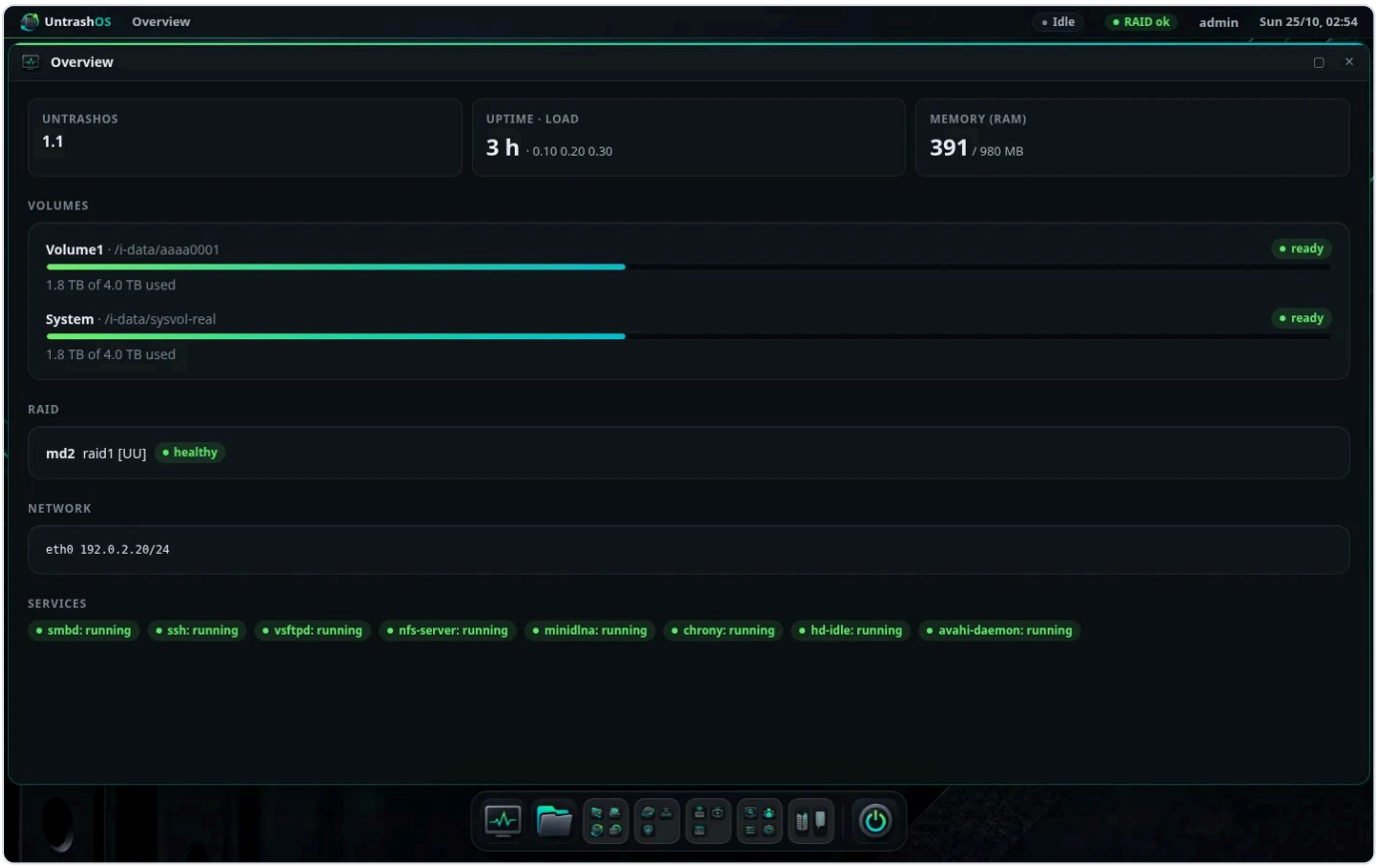


After way B or C, sign in. After way B, every user keeps the password from the Zyxel firmware. After way C, use `admin` and the password you chose in the installer.



Option	Meaning
Language	Language of this browser: English, Deutsch, Français.
User name, Password	Your account. Several wrong passwords in a row lock the sign-in for a while.
User guide	Opens this guide without signing in.

3. Working with the Desk

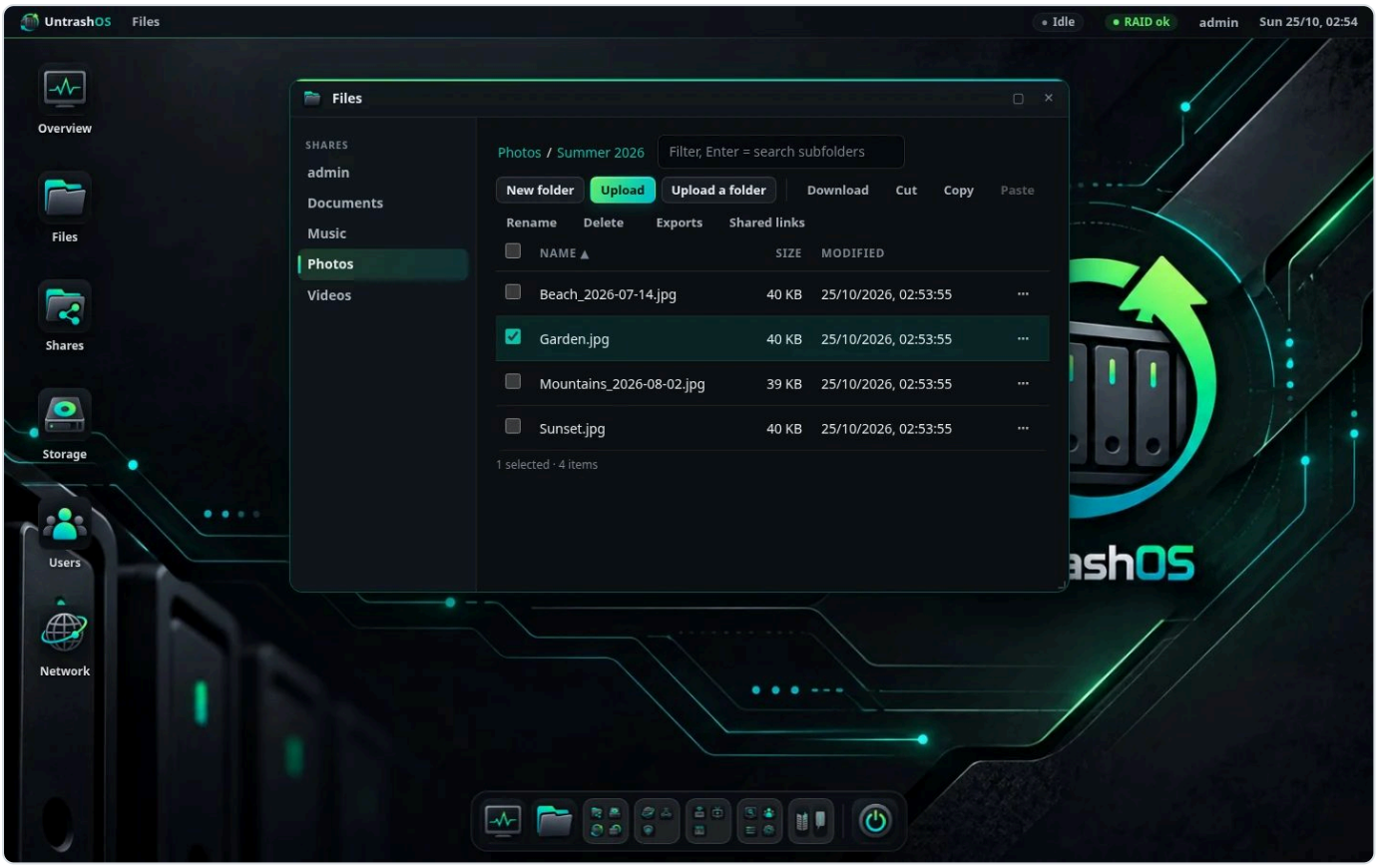


Element	Meaning
Top bar	Name of the active window; current activity (for example a media scan, streams, a running backup or update); **RAID** state; your **user name** (click it to change the language of this browser or to sign out); date and time.
Desktop icons	The first apps, opened with a click.
Dock	All apps. **Overview** and **Files** sit directly in the dock; the other apps are grouped in **Data**, **Network**, **Media**, **System** and **Help**: a click on a group opens its apps. Hover over an icon to see its name. The **power button** at the right end switches the NAS off or restarts it (section 16).
Windows	Move them by the title bar and resize them at the edges and corners. ▢ maximizes, and so does a double click on the title bar. ✕ closes. Each window keeps its size in your browser.

Overview shows the installed version, uptime and load, the memory in use, each volume with its fill level, the RAID arrays and their health, the network addresses and the state of the main services (SMB, SSH, FTP, NFS, media server, time, disk spin-down, mDNS).

Administrators see all apps. Normal users see **Files**, **Channels**, **Manual** and **My account** (section 17).

4. Files



Pick a share on the left. The path above the list leads back. Shares whose volume is missing are listed under *Volume missing* and cannot be opened.

Selecting. Click a row to select it; **Ctrl**+click adds or removes one, **Shift**+click selects a range, the box in the header selects all. Click a name (or double-click the row) to open it. The column headers sort by name, size or date.

Button	Meaning
New folder	Creates a folder in the current folder.
Upload / **Upload a folder**	Uploads files, or a whole folder with its subfolders, into the current folder. You can also drag files and folders from your computer onto the list (or onto a folder in it). The progress shows below the list.
Download	One file as it is; a folder or several items as one ZIP file.
Cut, **Copy**, **Paste**	Copies or moves the selection, also into another share: open the target folder and choose **Paste**. When names already exist you choose: keep both (the new ones get a number, "name (2)"), replace, or skip. Large copies run on the NAS in the background; the progress shows below the list and in the **Transfers** window (see below).
Rename	Renames the selected item.

Button	Meaning
Delete	Deletes the selection after a confirmation. On a share with a recycle bin the items go to its folder `recycle` (as for deletions over the network); inside `recycle`, or without a recycle bin, they are deleted permanently.
Filter field	Filters the current folder while you type; **Enter** searches the subfolders too. A click on a hit opens its folder.
Exports	Running and finished exports.
Shared links	Your download links with their expiry: **Copy** the address again or **Delete** the link. Administrators see the links of all users.

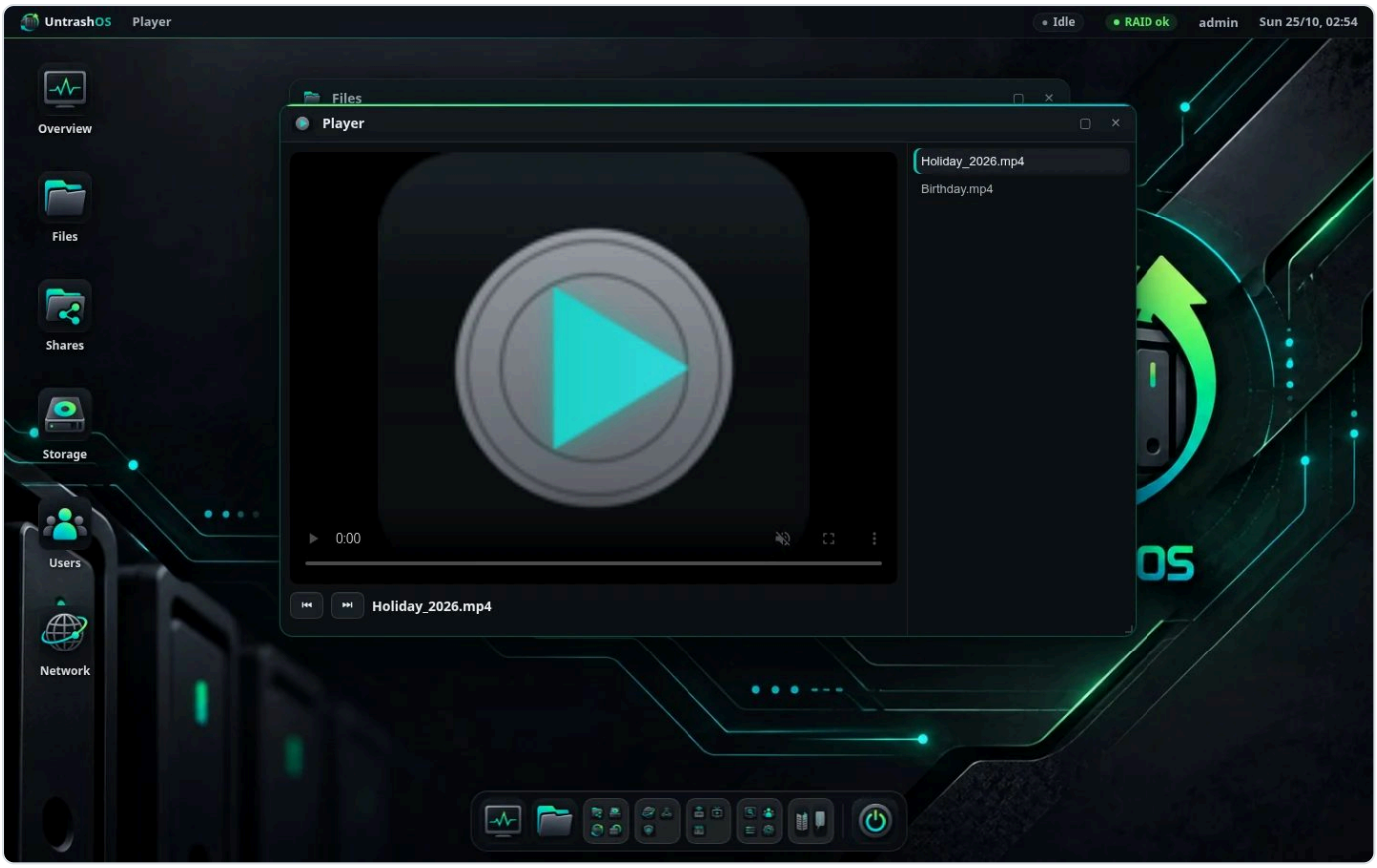
More (**⋯** at the end of a row, or a right click) has the same actions plus **Share link…** (a download link, see below), **Export…** (copy or convert, see below), **As share…** (folders, admins only: a share of its own, section 5) and **Properties** (with the size of a folder). Drag items onto a folder (or onto a part of the path above the list) to move them there; hold **Ctrl** to copy instead.

Transfers. The **Transfers** window (in the dock; for administrators under **Data**) lists every copy, move, upload and download of the Desk with its progress, speed and the time left. While something runs, the top bar shows how many transfers are running; a click opens the window. **Pause** holds a copy or a move where it is and **Resume** continues it; **Cancel** ends a transfer and removes the partly written file. Finished transfers stay in the list for an hour; × or **Finished ones from the list** remove them earlier. Users see their own transfers, administrators those of all users and also the jobs of the download service. Copies run as a process of their own, so the Desk stays responsive while a large copy runs, also onto a partner volume.

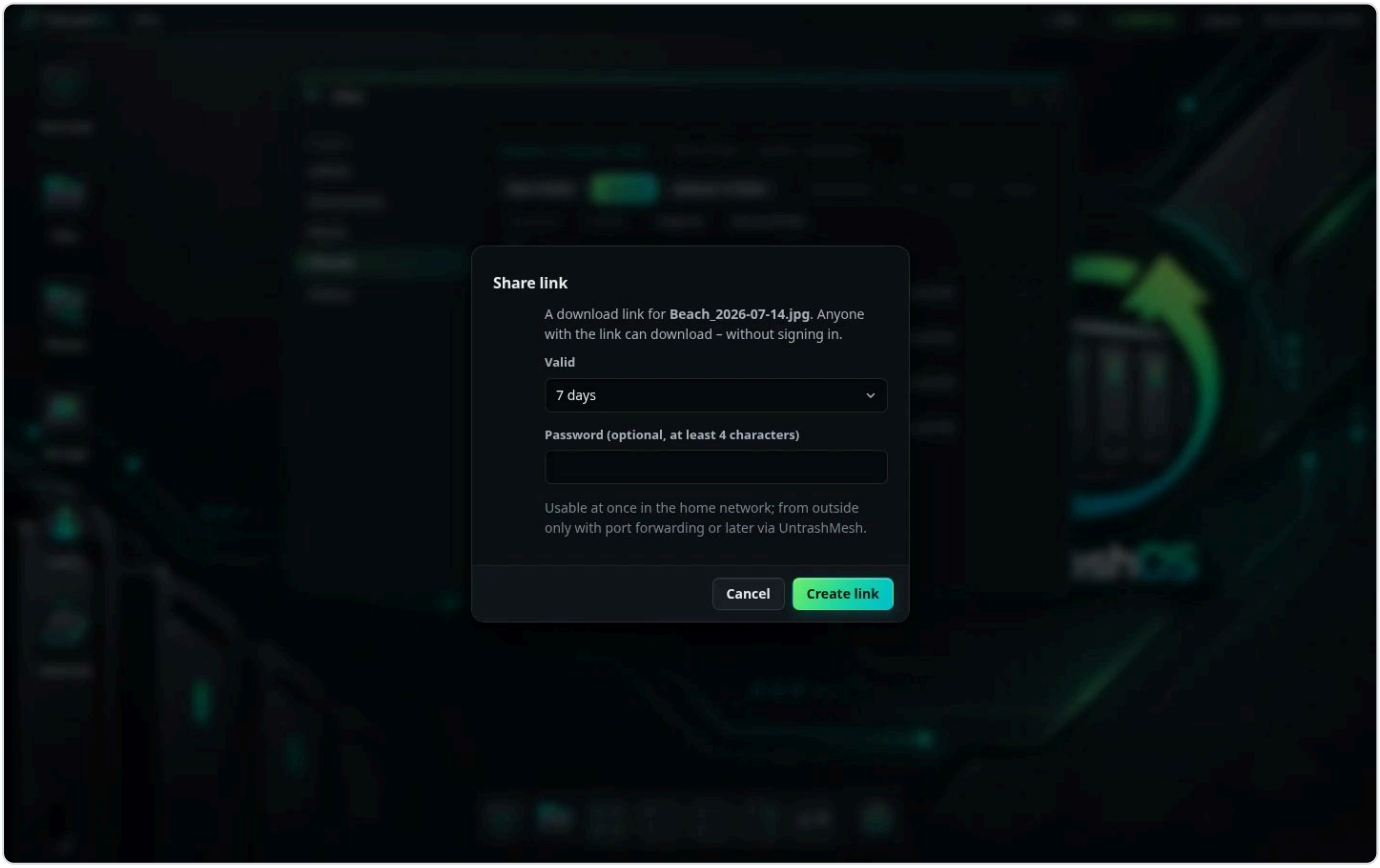
Key	Action
Ctrl+A, Ctrl+C, Ctrl+X, Ctrl+V	select all, copy, cut, paste
Delete, F2, Enter	delete, rename, open
Backspace, Escape, ↑ ↓	up one folder, clear the selection, move the selection

Clicking a file opens it:

File type	Opens in
Photos (JPEG, PNG, GIF, WebP, BMP)	**Photo viewer** with *Back*, *Next*, *Download*.
Video and music (MP4, MKV, AVI, MP3, FLAC, …)	**Player**, which plays every media file of the folder as a playlist. For videos with several audio tracks you can choose the track.
PDF and text files	**Document viewer** with page navigation and zoom.
Everything else	Download.


UntrashOS
Player
Idle
RAID ok
admin
Sun 25/10, 02:54
Overview
Files
Shares
Storage
Users
Network
Files
Player
Holiday_2026.mp4
Birthday.mp4
0:00
Holiday_2026.mp4

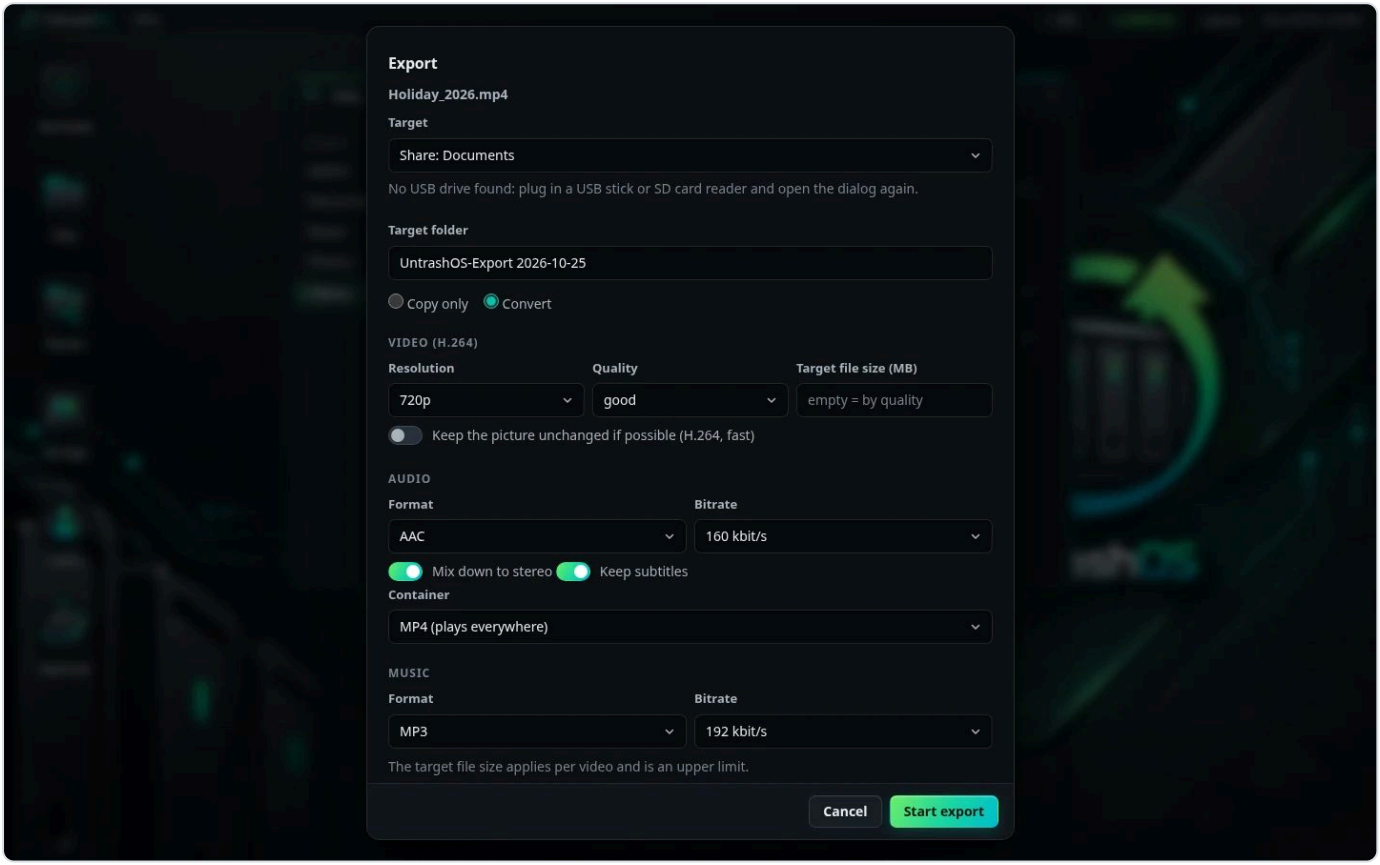
Share link


Share link
A download link for Beach_2026-07-14.jpg. Anyone with the link can download – without signing in.
Valid
7 days
Password (optional, at least 4 characters)
Usable at once in the home network; from outside only with port forwarding or later via UntrashMesh.
Cancel
Create link

Option	Choices	Meaning
Valid	1 day, **7 days**, 30 days, 90 days	When the link stops working.
Password	optional, at least 4 characters	Asked before the download.

Anyone with the link can download the file, or a folder as ZIP, without signing in. Links work in the home network at once. From outside they need a port forwarding (section 8).

Export

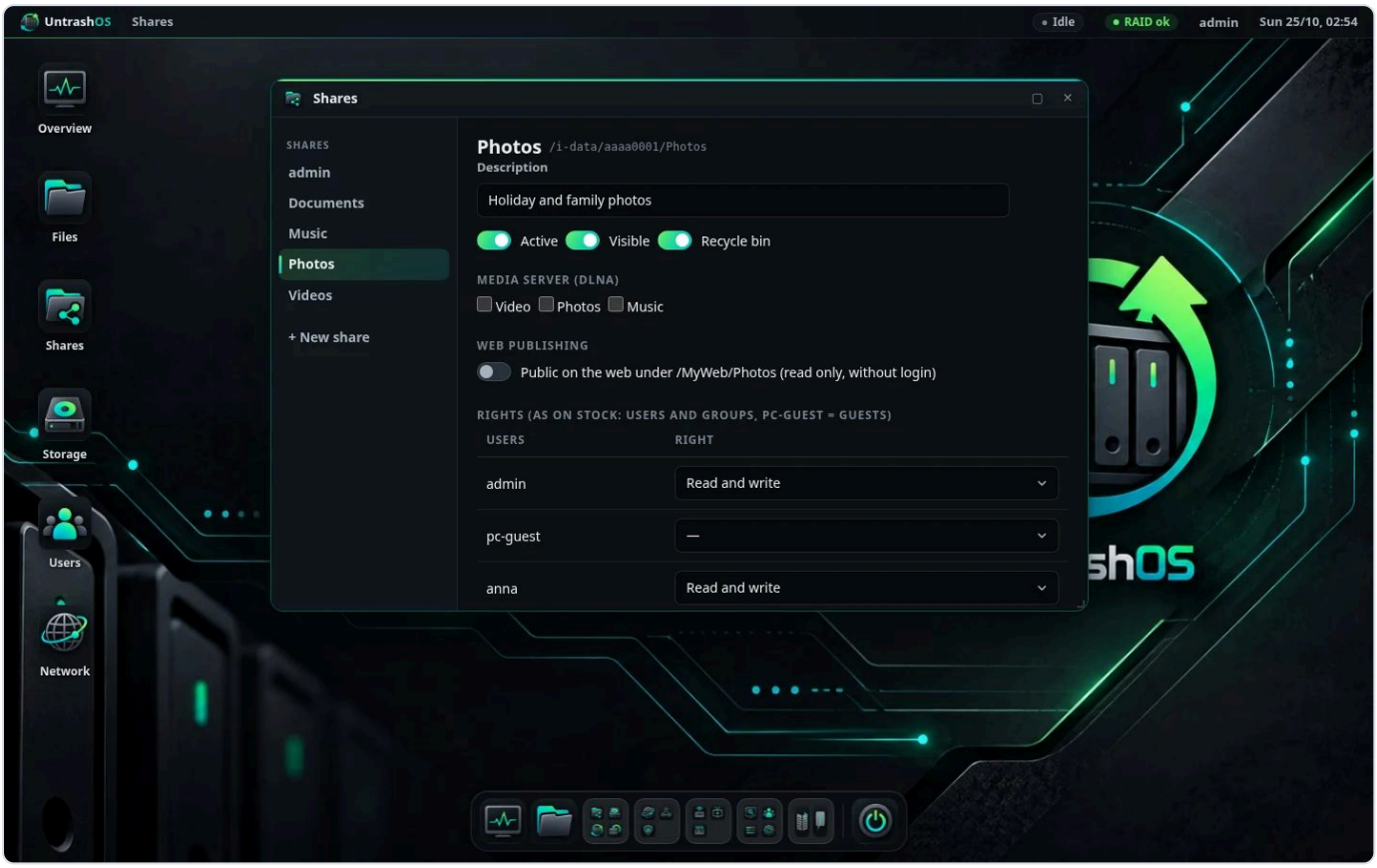


Option	Choices	Meaning
Target	**USB/SD drives**, then every share	Where the copy goes. Plug in a USB stick or SD card reader and reopen the dialog to see it.
Target folder	default **UntrashOS-Export**	Folder created at the target.
Mode	**Copy only**, Convert	*Copy only* copies 1:1. *Convert* recodes video and music as set below; other files are copied.
Video: Resolution	as original, 1080p, **720p**, 480p, 360p	Maximum picture height.
Video: Quality	high, **good**, medium, small	Picture quality versus file size.

Option	Choices	Meaning
Video: Target file size (MB)	empty = by quality	Upper limit per video; overrides the quality.
Keep the picture unchanged if possible	**off**, on	Keeps H.264 video as it is and only converts the sound (fast).
Audio: Format	**AAC**, MP3, Opus, unchanged	Sound of videos.
Audio: Bitrate	96, 128, **160**, 192, 256 kbit/s	Sound quality.
Mix down to stereo	**on**, off	Turns surround sound into stereo.
Keep subtitles	**on**, off	Keeps subtitle tracks.
Container	**MP4 (plays everywhere)**, MKV (all subtitles)	File format of the video.
Music: Format	**MP3**, AAC, Opus, FLAC, unchanged	Format for music files.
Music: Bitrate	128, **192**, 256, 320 kbit/s	Quality for music.

Converting runs in the background and is slow on the NAS, which has no hardware encoder: a feature film in 720p takes several hours. The top bar and **Exports** show the progress.

5. Shares

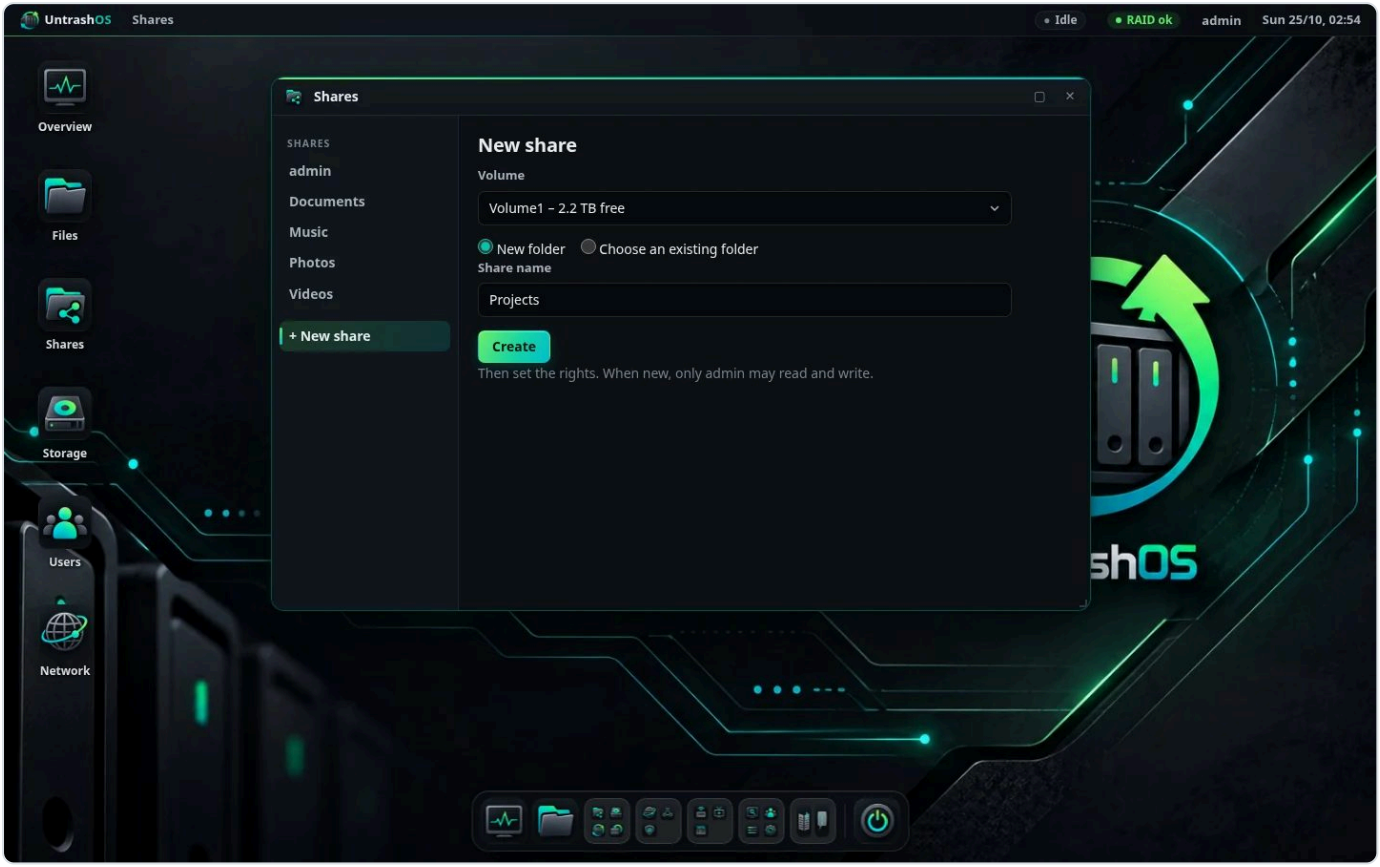


Choose a share on the left. Disabled shares are marked *(off)*.

Option	Choices	Meaning
Description	text	Shown in the network.
Active	**on**, off	Off hides the share from the network; the folder stays.
Visible	**on**, off	Off hides the share from network browsing; it can still be opened by name.
Recycle bin	on, off	Deleted files go to a `recycle` folder of the share (section 9, *Recycle bin*).
Media server (DLNA)	Video, Photos, Music	Which media types the media server publishes from this share.
Web Publishing	**off**, on	Readable on the web under `/MyWeb/<share>`, without login and read only. Needs *Web Publishing* in Settings → Services.
Rights per user and per group	—, Read and write, Read only, No access	As on the Zyxel firmware. `pc-guest` stands for guests. — means no rule of its own. *No access* wins over everything else.

Save applies the changes. **Remove share** removes the share from the network. Its folder and all files in it stay, as on the Zyxel firmware. The built-in share `admin` cannot be removed.

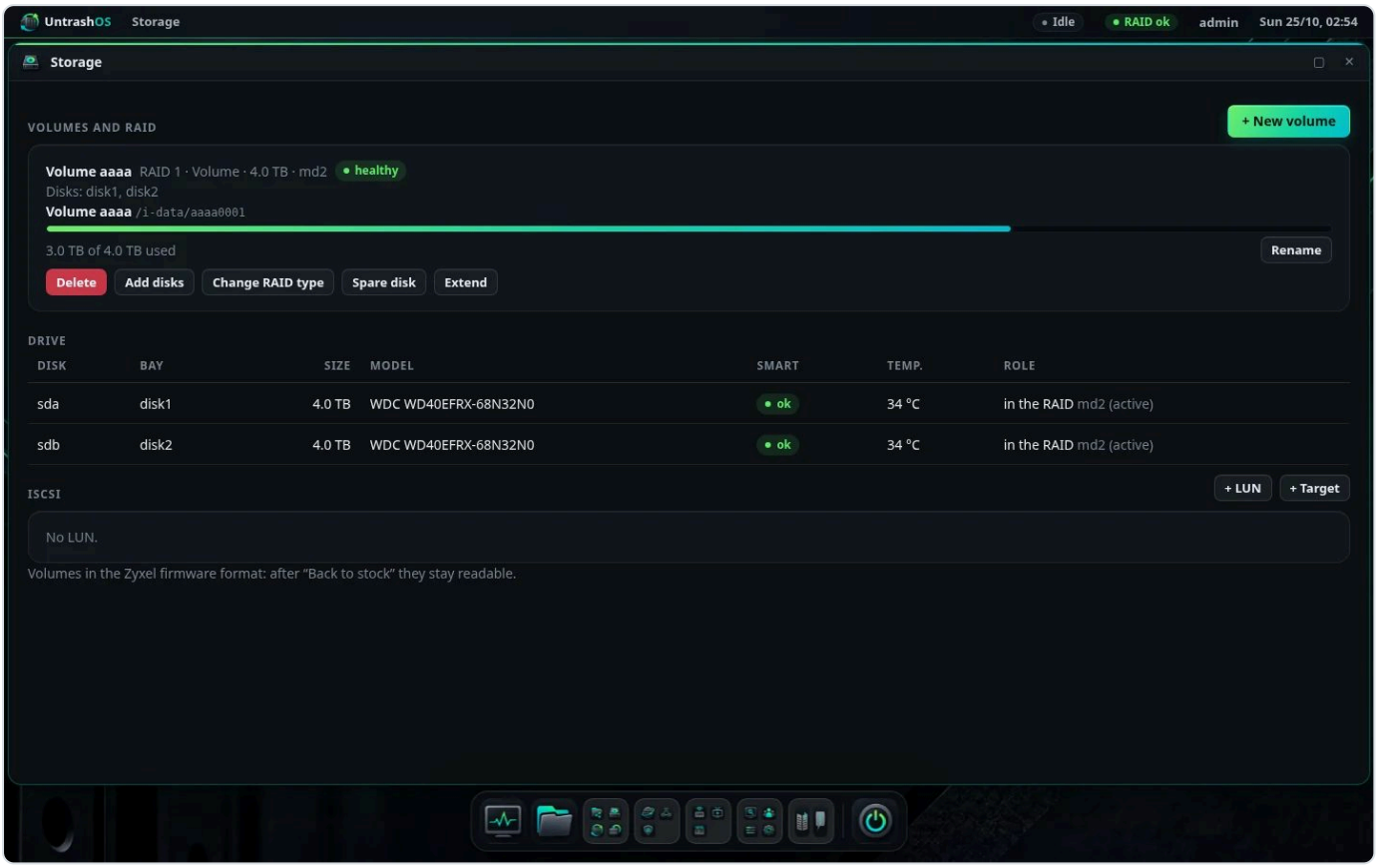
New share



Option	Choices	Meaning
Volume	the internal volumes with free space	Where the share lives.
Kind	**New folder**, Choose an existing folder	*Choose an existing folder* opens a folder browser; select the folder with **Choose this folder**.
Share name	text	Name in the network.

A new share is readable and writable only by `admin`. Set its rights afterwards.

6. Storage



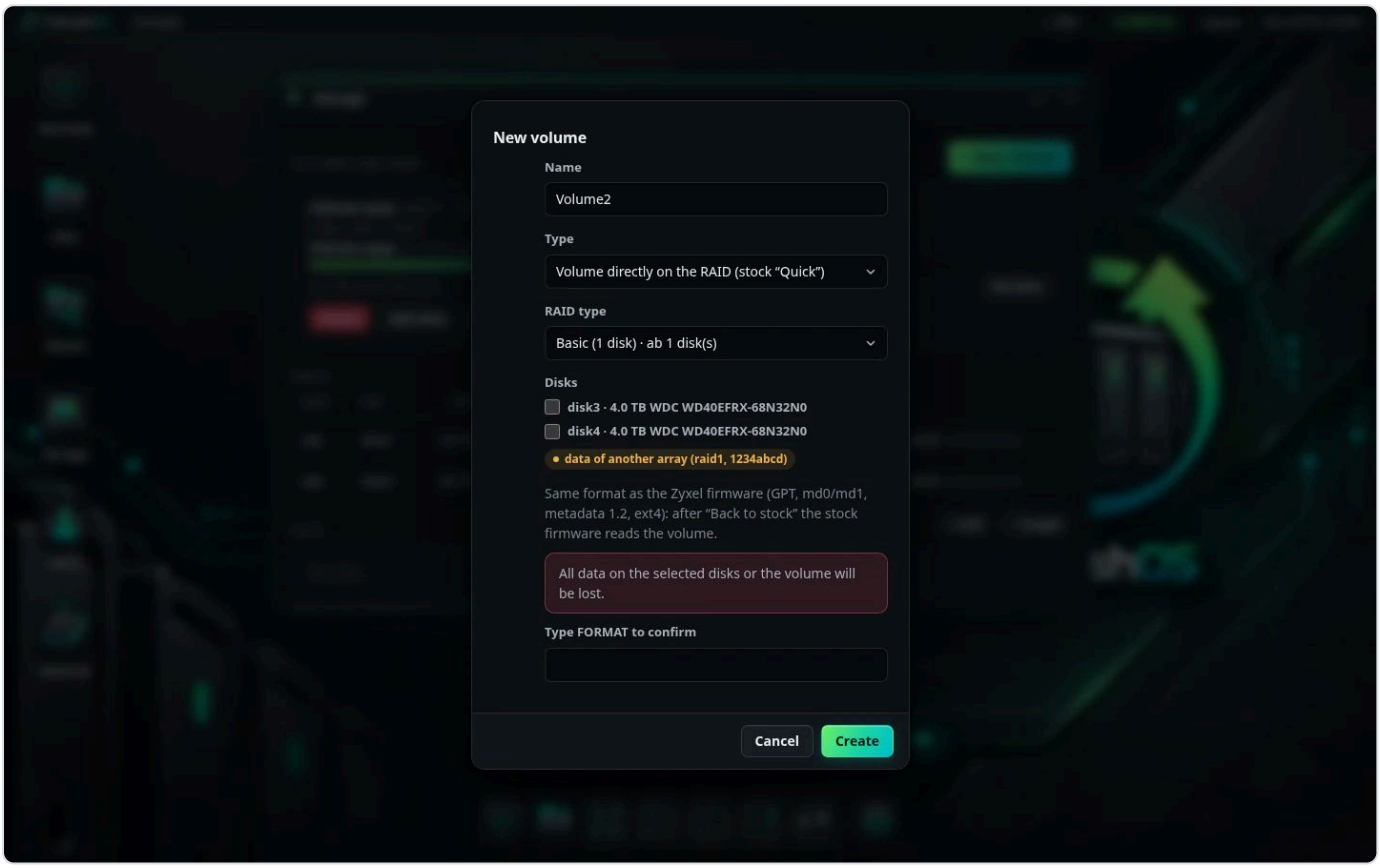
Volumes and RAID lists each array with its RAID type, size, disks, state (*healthy*, *degraded*, *resyncing*, *failed*) and its volumes with their fill level. **Drive** lists every disk with bay, size, model, SMART state, temperature and role (*in the RAID*, *free*, *free (old data)*, *UntrashOS*, *USB*). All volumes are created in the format of the Zyxel firmware, so the Zyxel firmware can still read them after the way back.

Action	What it does	Confirmation
+ New volume	New array on free disks, see below.	`FORMAT`
Rename	New name for the volume.	—
Add disks	Extends the array with free disks; the RAID type stays.	`FORMAT`
Change RAID type	Moves to the next RAID type (for example RAID 1 → RAID 5) with additional disks.	`FORMAT`
Spare disk	Adds a hot spare that takes over when a disk fails.	`FORMAT`
Repair (only for degraded arrays)	Rebuilds the array onto a free disk.	`FORMAT`
Extend	After swapping all disks for larger ones: grows the array and file system to the new size.	—

Action	What it does	Confirmation
Delete	Deletes the volume and every share on it.	`DELETE`

The confirmation word protects the disks: everything on the selected disks, or on the deleted volume, is lost. A running job is shown with its progress and log.

New volume



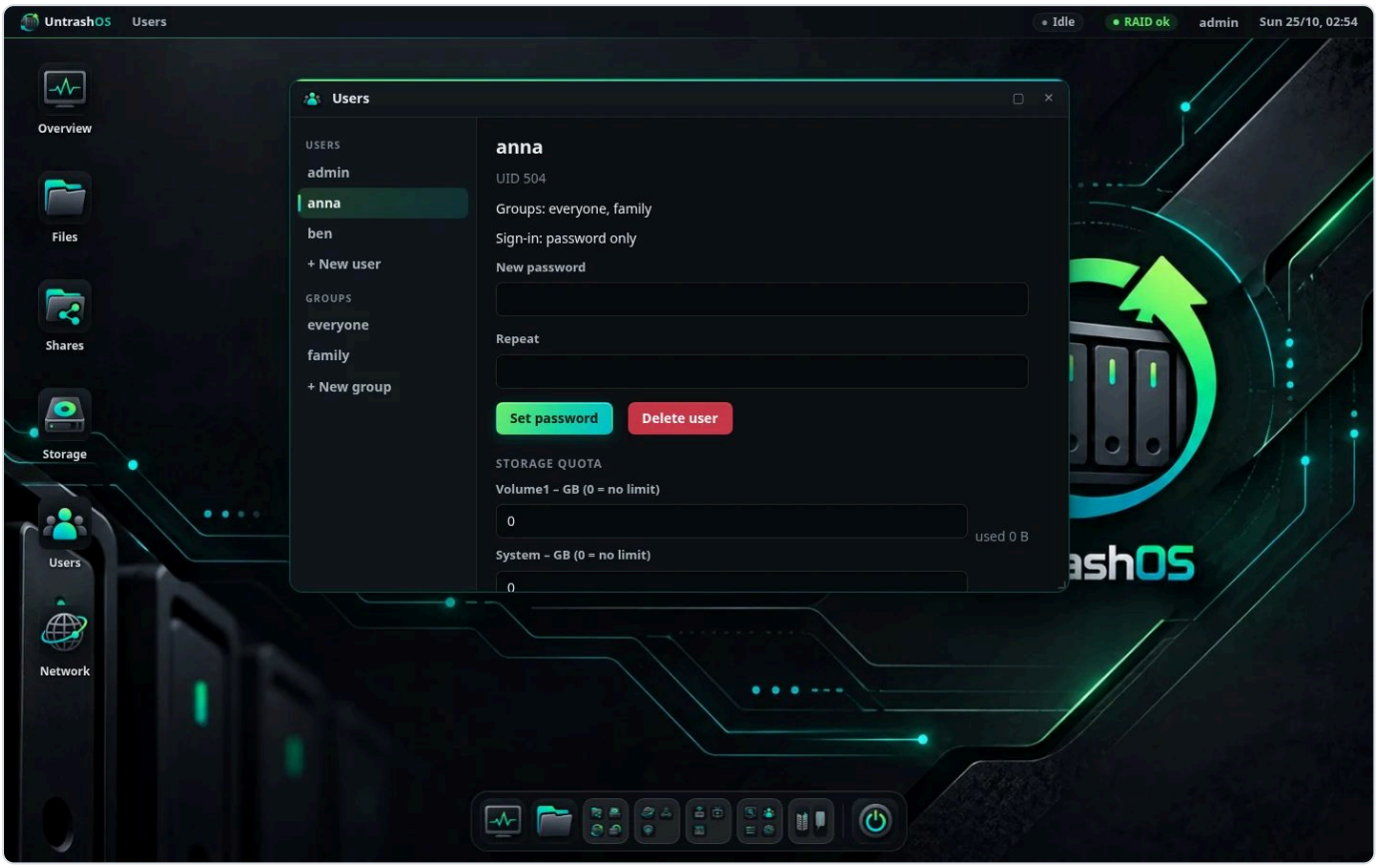
Option	Choices	Meaning
Name	default **Volume2**	Name of the volume.
Type	**Volume directly on the RAID** (Zyxel "Quick"), Disk group with volumes (Zyxel "Expert", LVM)	*Disk group* allows several volumes of chosen sizes on one array.
RAID type	**Basic** (1 disk), JBOD (from 2), RAID 0 (from 2), RAID 1 (from 2), RAID 5 (from 3), RAID 6 (from 4), RAID 10 (from 4)	Basic: one disk. JBOD: disks joined, no redundancy. RAID 0: striped, fast, no redundancy. RAID 1: mirrored. RAID 5: survives one failed disk. RAID 6: survives two. RAID 10: mirrored and striped.
Disks	the free disks	Which disks to use.
First volume: name, Size in GB	only for *Disk group*; empty size = everything	The first volume in the disk group.

iSCSI

Volumes can be offered to other computers as iSCSI disks.

Dialog	Option	Meaning
+ LUN	Name, Volume, Size in GB	A virtual disk in a file on a volume.
	Thin (**on**)	Space is taken only when it is used.
	Write cache (**off**)	Faster, but data in the cache is lost on a power failure.
+ Target	Name, IQN (empty = automatic)	The address computers connect to.
	Only these initiators	Comma-separated IQNs that may connect; empty = all.
	CHAP, CHAP user name, CHAP password (12–16 characters)	Login required to connect. An empty password keeps the old one.
	Mutual CHAP, user name, password	The NAS also proves itself to the computer.
	Active (**on**)	Offers the target.

7. Users and groups

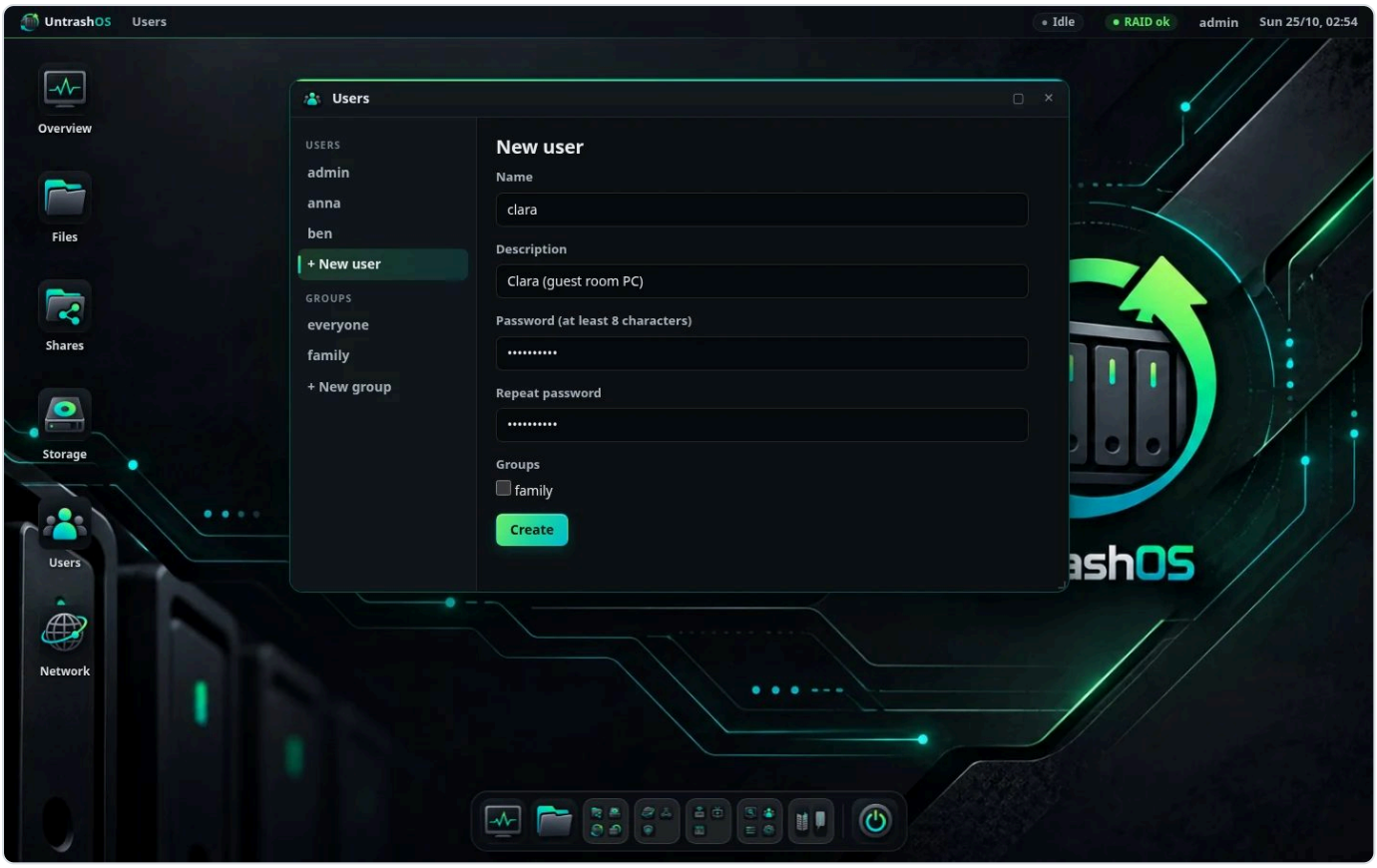


User page

Option	Meaning
New password, Repeat	Sets the user's password (Desk, SMB, FTP, SSH for admin).
Delete user	Removes the account. `admin` cannot be deleted.
Storage quota	Per volume, in GB; **0** = no limit. The page shows how much the user already uses.

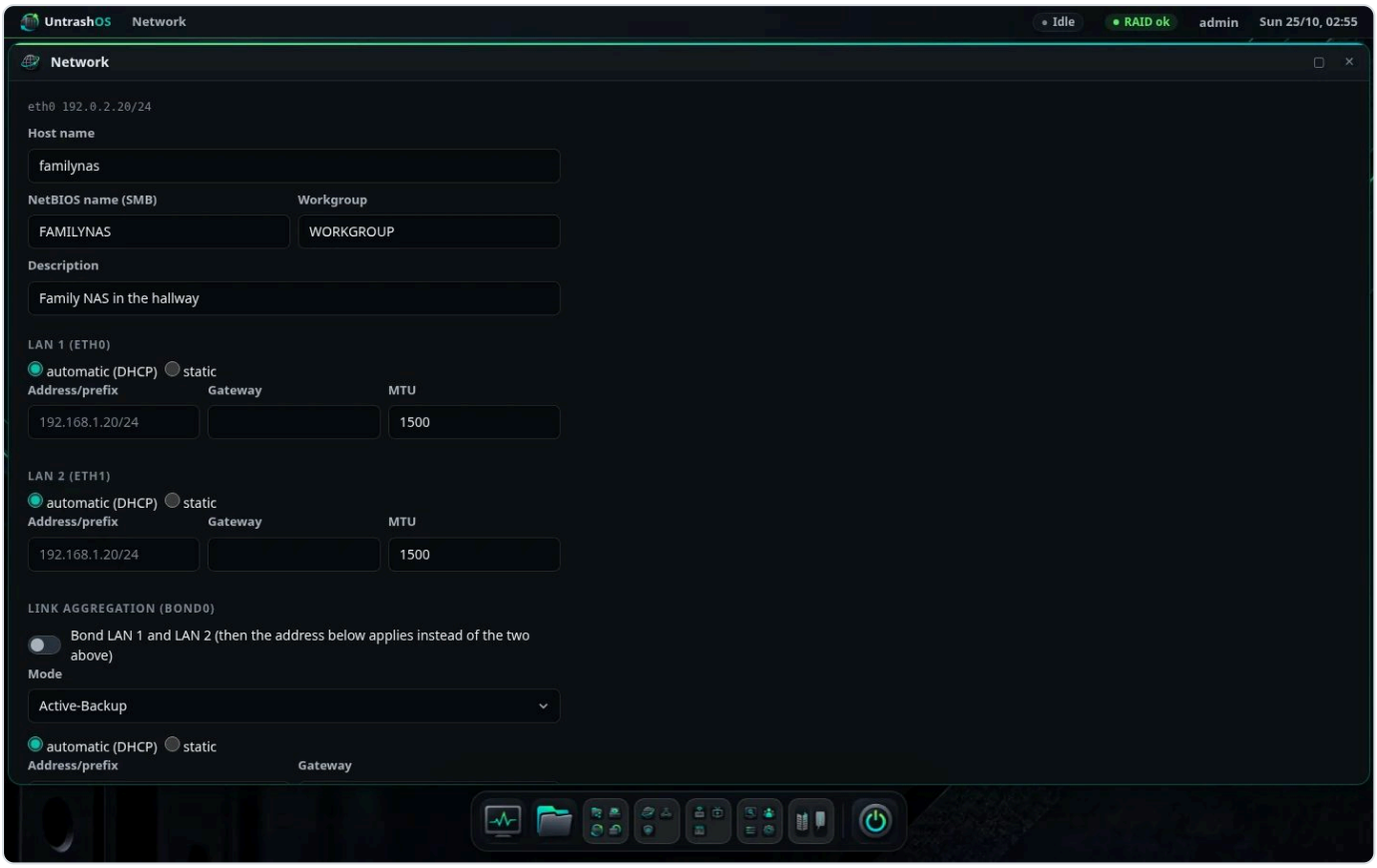
+ New user

Option	Meaning
Name	Account name.
Description	Optional, for example the full name.
Password, Repeat password	At least 8 characters.
Groups	Groups the user joins. Every user is also in `everyone`.



Groups: choose a group to tick its members and **Save**, or **Delete group**. **+ New group** asks for a name and a description. Groups are used for share rights.

8. Network



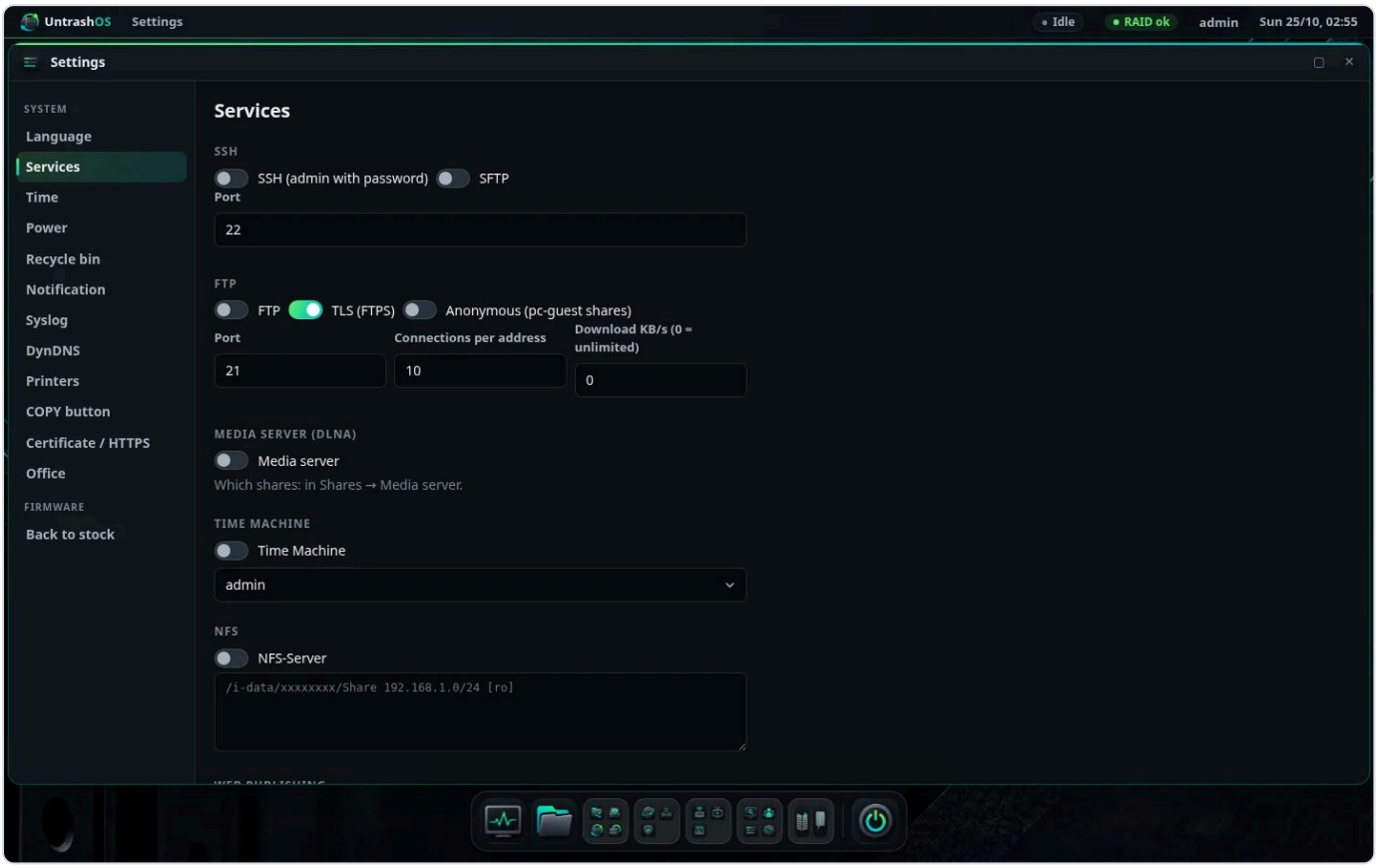
Option	Choices	Meaning
Host name	default **untrash** (`untrash-2` , … if taken)	Name of the NAS in the network (`<host name>.local`).
NetBIOS name (SMB), Workgroup, Description	text	Name, workgroup and comment for Windows networking.
LAN 1 (eth0), LAN 2 (eth1)	**automatic (DHCP)**, static	With *static*: Address/prefix (address and prefix length, for example `<LAN_IP>/24`), Gateway. MTU default **1500**.
Link aggregation (bond0)	**off**, on	Bonds both ports. The address of the bond then applies instead of the two above.
Bond mode	Round-Robin, **Active-Backup**, XOR, Broadcast, 802.3ad (LACP, switch required), Adaptive transmit load balancing, Adaptive load balancing	*Active-Backup* works with every switch and gives fail-over. *802.3ad* needs a switch that supports LACP.

Option	Choices	Meaning
DNS	**from the DHCP server**, or DNS 1 and DNS 2	Name servers of the NAS.

Save and apply activates the settings. A wrong static address makes the NAS unreachable. The way back: connect the NAS directly to a PC with a network cable. It is then reachable by its automatic address as `<host name>.local`.

Router (UPnP port forwarding) opens ports on the router for access from the internet. For each service enter a WAN port (empty = off): Untrash Desk (HTTP), Untrash Desk (HTTPS), FTP, SSH, WebDAV, WebDAV (HTTPS), Web Publishing, media server. The table shows what the router currently forwards. **Forwarded services are reachable from the internet.** Use HTTPS and strong passwords.

9. Settings



Language

Option	Choices	Meaning
System language	**English**, Deutsch, Français	For everyone who has not chosen a language in their browser.
Language of this browser	**— as system —**, English, Deutsch, Français	Only for the browser you are using.

Services

Option	Choices	Meaning
SSH (admin with password)	**off**, on	Shell access for `admin`. Port default **22**.
SFTP	**off**, on	File transfer over SSH.
FTP	**off**, on	FTP server. Port **21**.
TLS (FTPS)	**on**, off	Encrypted FTP. Uses the certificate from *Certificate / HTTPS*.

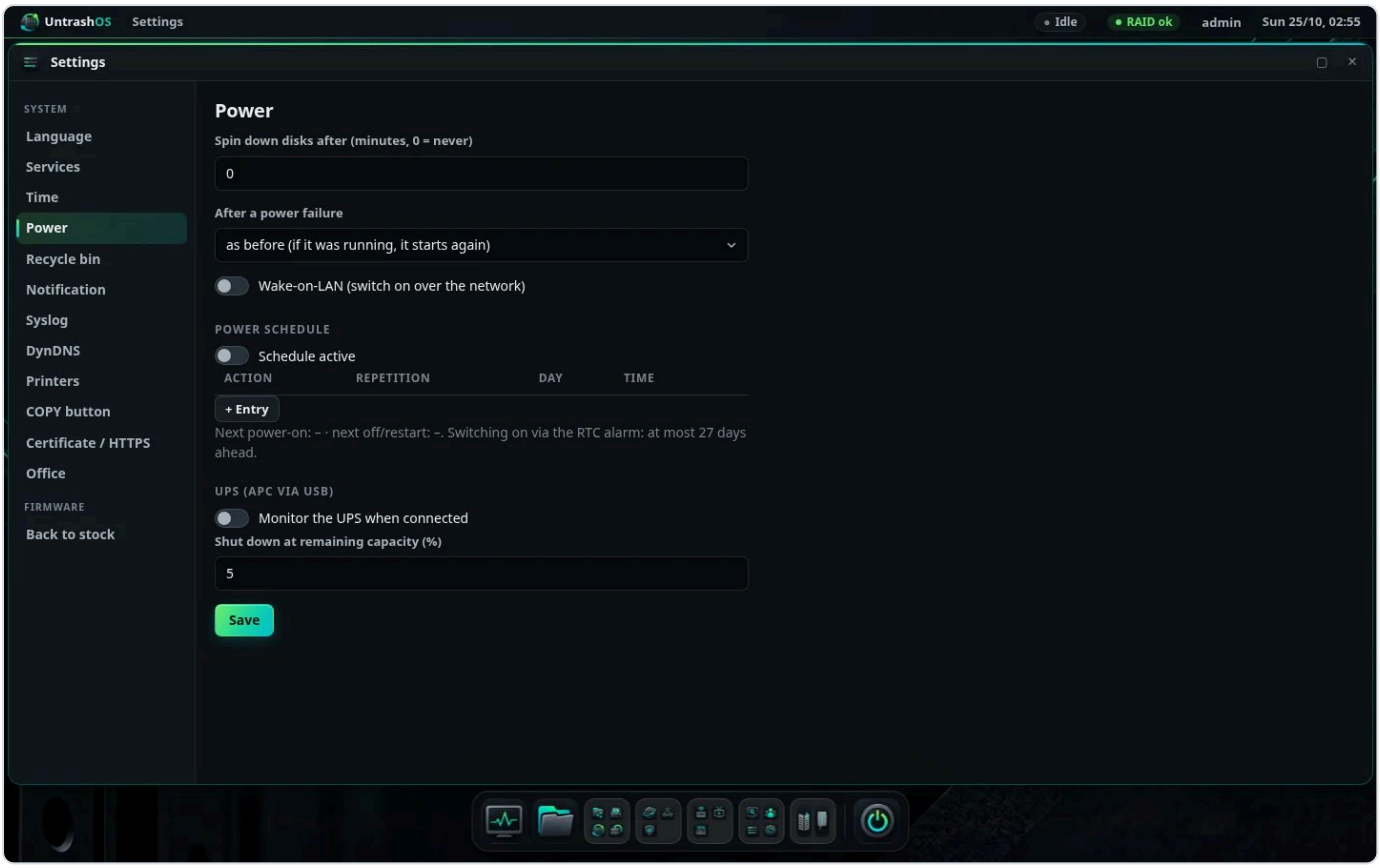
Option	Choices	Meaning
Anonymous (pc-guest shares)	**off**, on	FTP without login for the guest shares.
Connections per address	default **10**	Limit per client.
Download KB/s	**0** = unlimited	Speed limit for FTP downloads.
Media server	**off**, on	DLNA/UPnP server (section 11).
Time Machine	**off**, on, with a share	Backup target for Macs.
NFS server	**off**, on	One export per line: `/i-data/<volume>/<share> <network>/<prefix> [ro]`.
Web Publishing	**off**, on; Port **80**	Shares marked for Web Publishing under `http://<NAS>/MyWeb/<share>`.
WebDAV	**off**, on; Anonymous (guest shares) **off**	`http://<NAS>:5002/webdav/<share>`, HTTPS on port 5003. Same rights as the share.

Save and apply restarts the changed services.

Time

Option	Choices	Meaning
Time zone	all time zones, default **Europe/Berlin**	Local time of the NAS.
Time from the network (NTP)	**on**, off	Sets the clock automatically. NTP server default **pool.ntp.org**.

Power

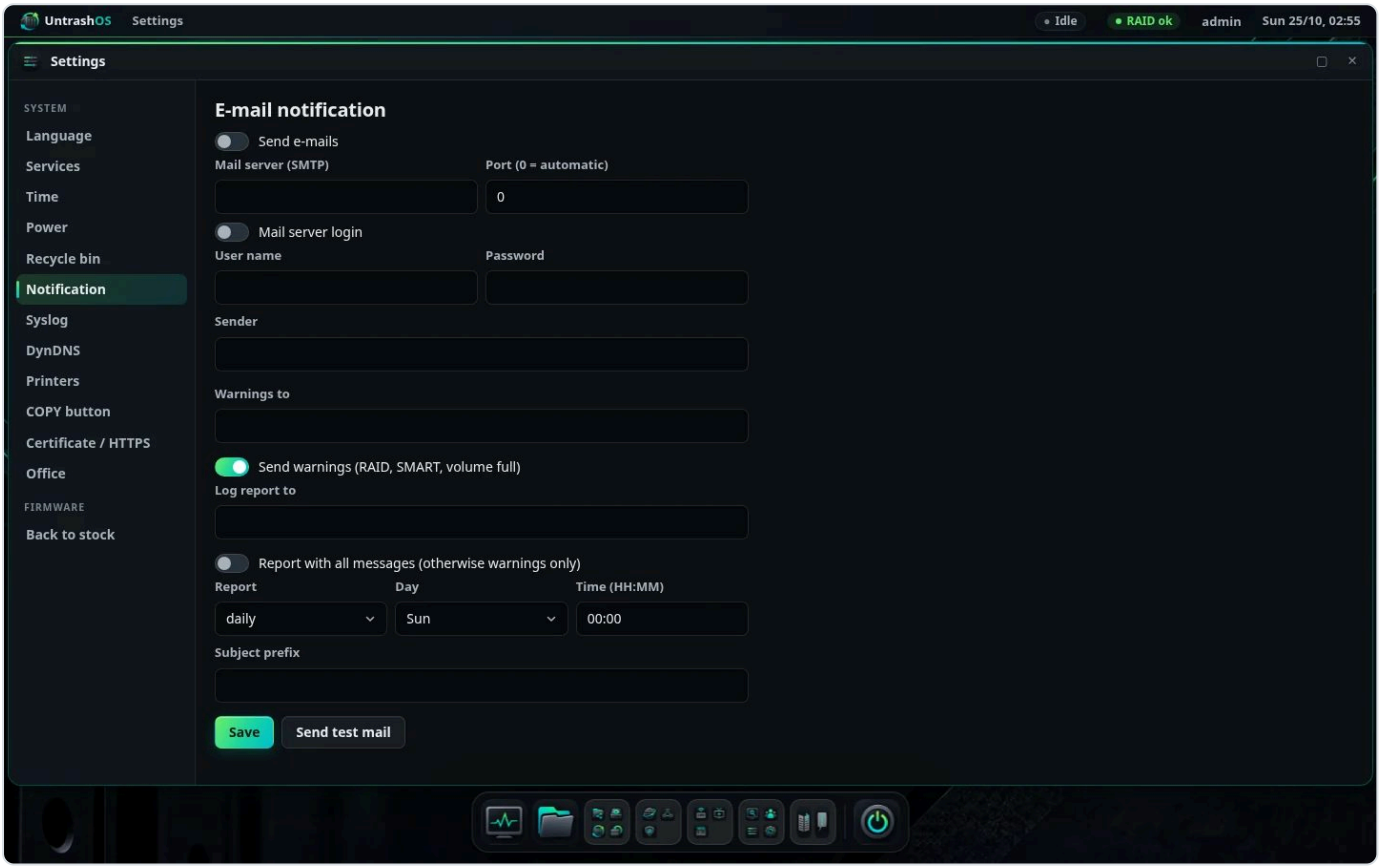


Option	Choices	Meaning
Spin down disks after	minutes, **0** = never	Idle disks stop spinning.
After a power failure	**as before** (if it was running, it starts again), always switch on, stay off	Behaviour when the power returns.
Wake-on-LAN	**off**, on	Switch the NAS on with a network packet.
Power schedule	**off**, on; entries with Action (switch on, **switch off**, restart), Repetition (**daily**, weekly, monthly), Day, Time	Switches the NAS on and off by the clock. The page shows the next planned events. Switching on uses the alarm clock of the NAS, which reaches at most 27 days ahead.
Monitor the UPS when connected	**off**, on	For an APC UPS on USB.
Shut down at remaining capacity	% , default **5**	The NAS shuts down cleanly when the UPS battery reaches this level.

Recycle bin

Option	Choices	Meaning
Delete permanently after days	**0** = never	Files in the `recycle` folders are deleted after this many days. Shares need *Recycle bin* switched on (section 5).

Notification



Option	Choices	Meaning
Send e-mails	**off**, on	Switches e-mail sending on.
Mail server (SMTP), Port	Port **0** = automatic	Your provider's outgoing mail server.
Mail server login	**off**, on; User name, Password	Most providers require a login.
Sender	address	From-address of the e-mails.
Warnings to	address	Receives warnings.
Send warnings (RAID, SMART, volume full)	**on**, off	Sends an e-mail at once when one of these occurs.
Log report to	address	Receives regular log reports.
Report with all messages	**off** (warnings only), on	Content of the report.

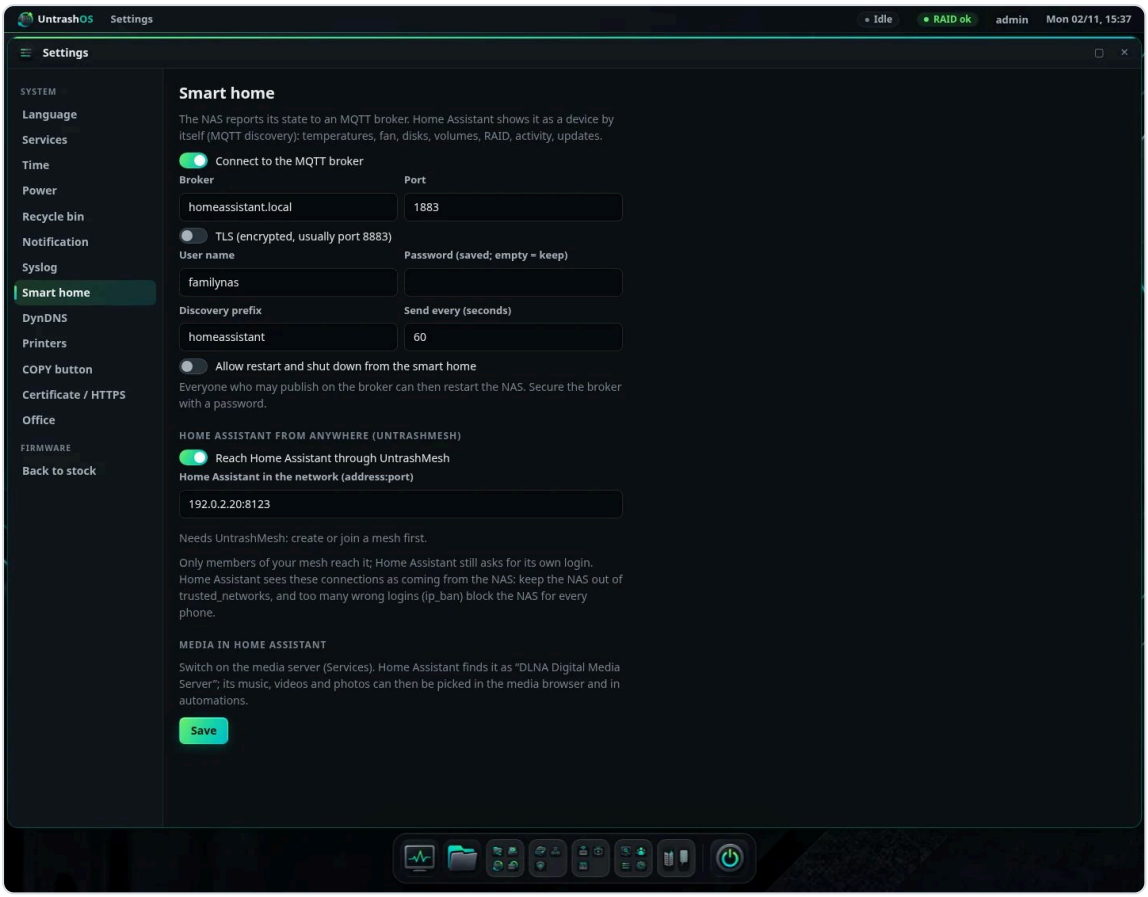
Option	Choices	Meaning
Report	hourly, **daily**, weekly; Day (for weekly); Time **00:00**	When the report is sent.
Subject prefix	text	Put in front of every subject, to sort the e-mails.

Send test mail checks the settings.

Syslog

Option	Choices	Meaning
Forward	**off**, on; Server, Port (UDP) **514**	Sends the NAS log to a syslog server.
Syslog server on port 514	**off**, on	Collects the logs of other devices on the NAS.
Share, Folder	default folder **syslog**	Where collected logs are stored.
Format	**simple** (date, device, message), complete	Line format.
Keep months	default **6**	Older logs are deleted.

Smart home



The NAS can show up as a device in a smart home that speaks MQTT, for example Home Assistant. Home Assistant itself does not run on the NAS (it no longer supports 32-bit ARM); it runs on its own device, and the NAS reports to it.

Option	Choices	Meaning
Connect to the MQTT broker	**off**, on	The NAS sends its state to the broker.
Broker, Port	address; **1883** (8883 with TLS)	The MQTT broker, for example the Mosquitto add-on of Home Assistant.
TLS (encrypted, usually port 8883)	**off**, on	Encrypted connection; the broker needs a certificate that the NAS trusts.
User name, Password	text	The broker account. An empty password field keeps the saved password.
Discovery prefix	**homeassistant**	Where the device description goes. Change it only if your Home Assistant uses another prefix.
Send every (seconds)	**60** (10–3600)	How often the values are sent.

Option	Choices	Meaning
Allow restart and shut down from the smart home	**off**, on	Adds the buttons *Restart* and *Shut down*. Everyone who may publish on the broker can then use them, so secure the broker with a password.

What Home Assistant shows. Home Assistant finds the NAS by itself (MQTT discovery) as a device named after the NAS. It shows:

- CPU, memory, swap and load
- board temperature and fan level
- per bay, the disk temperature and whether the disk sleeps; a sleeping disk is never woken for this
- per volume, the space used and free
- per disk group, a problem flag when it is degraded, and the progress of a rebuild or check
- what the NAS is doing (streams, devices on the shares, backup, …)
- the last start, and an update entry with the installed and the newest version

Values of a disk or volume that is removed disappear from Home Assistant as well.

Home Assistant from anywhere. With UntrashMesh, the NAS can pass Home Assistant on to the devices of your mesh. Switch on **Reach Home Assistant through UntrashMesh** and enter the address of Home Assistant in your home network as `<IP address>:8123` (a host name works too, but an IP address is safer). On a phone with the UntrashApp connected, enter the address that the page shows, `http://[<mesh address of the NAS>]:8123`, in the Home Assistant app (for example as its external address). The port is open only inside the mesh, not in your home network or on the internet. From outside your home the phone must reach the NAS: see UntrashMesh in Status, logs and system tools (UPnP or UDP port 51820).

Notes for Home Assistant:

- Home Assistant sees these connections as coming from the NAS. Do not add the NAS to `trusted_networks`; otherwise anyone in the mesh is signed in without a password.
- If Home Assistant bans addresses after wrong logins (`ip_ban`), a ban of the NAS address blocks every phone that comes in through the mesh. Remove the NAS from `ip_bans.yaml` to lift it.
- All members of the mesh reach the port, partner NAS too; Home Assistant still asks for its own login.

Media in Home Assistant. Switch on the media server. Home Assistant offers it through its *DLNA Digital Media Server* integration. The music, videos and photos of the published shares can then be picked in the media browser of Home Assistant and in scenes and automations (for example with the action *Media player: Play media*). The player plays the files as they are, so it must support their format.

DynDNS

Option	Choices	Meaning
Keep the address current at the DynDNS provider	**off**, on	Updates a host name when your internet address changes.
Provider	**DynDNS (dyn.com)**, No-IP, 3322.org, ZoneEdit, Duck DNS, FreeDNS (afraid.org), dynv6	Your DynDNS service.
Host name, User name, Password / token	text	Your account at the provider.

Printers

Option	Choices	Meaning
Share USB printers on the network (SMB and IPP)	**on**, off	Connected USB printers get a queue without a driver; the computer that prints brings the driver. The page lists known printers and their jobs.

COPY button

The COPY/SYNC button on the front of the NAS works with the most recently connected USB drive. **Short press = copy, hold 3 seconds = synchronize.**

Option	Choices	Meaning
Copy: NAS share	**– none –**, every share	Share used for copying; *none* switches the short press off.
Copy: Direction	**USB → NAS**, NAS → USB	
Keep replaced files in	**– do not keep –**, every share	Files that a copy would overwrite are kept there.
Copy into a new (dated) folder	**on**, off	Every copy goes into its own folder named by date.
Synchronize: NAS share	**– none –**, every share	Share used for synchronizing; *none* switches the long press off.
Synchronize: Direction	**both directions**, NAS → USB (mirror), USB → NAS (mirror)	*Mirror* makes the target equal to the source and deletes what is not in the source.
Keep replaced/deleted files in	**– do not keep –**, every share	Safety copy of overwritten or deleted files.

Copy now and **Synchronize now** do the same as the button.

Certificate / HTTPS

Option	Meaning
Name (CN), Create a self-signed certificate	Creates a certificate for the given name. Browsers warn about self-signed certificates once.
Own certificate	Paste your certificate (with intermediate certificates) and the unencrypted private key, then **Upload**.
HTTPS (**off**), Port **443**	Serves the Desk over HTTPS.
Redirect HTTP to HTTPS (**off**)	Sends every browser to HTTPS.

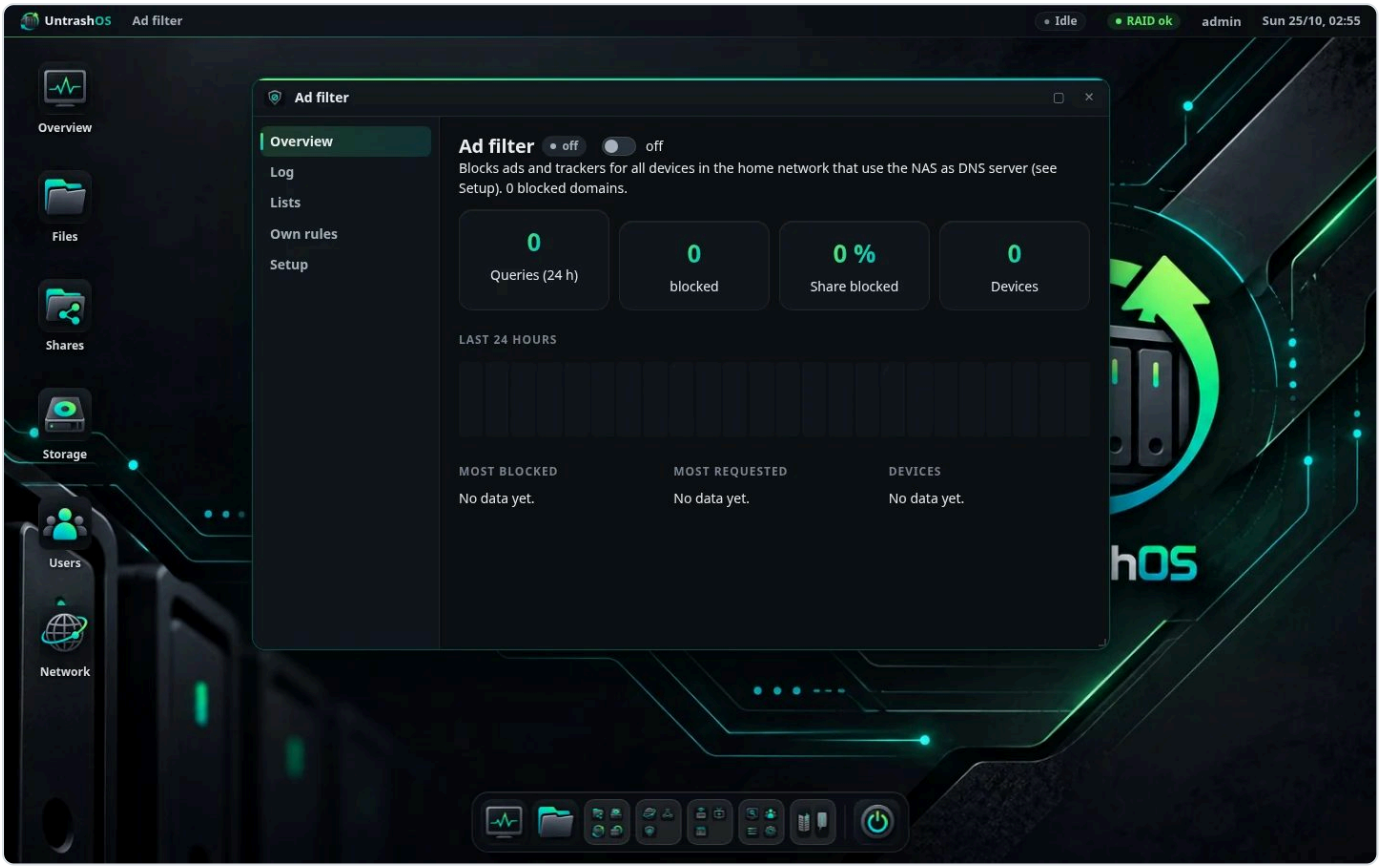
The certificate is also used for FTPS and WebDAV over HTTPS. **Save** restarts the Desk.

Back to stock

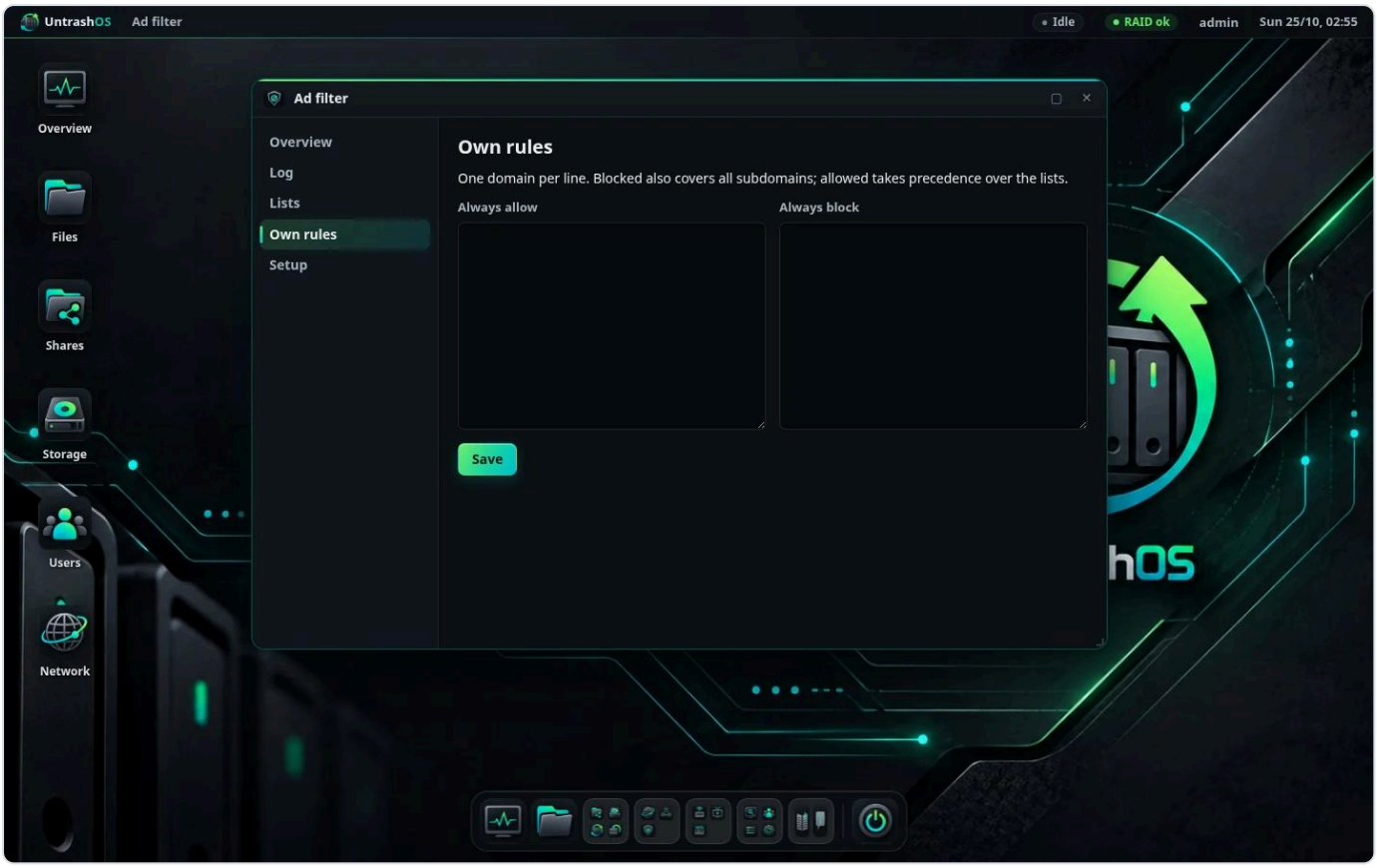
See section 18.

10. Ad filter

The ad filter blocks ads and trackers for every device in the home network that uses the NAS as its DNS server.



Page	Content and options
Overview	Switch **on/off**; queries and blocked queries in the last 24 hours, share blocked, devices; a chart of the last 24 hours; most blocked and most requested domains; the devices.
Log	Single queries. *Filter by domain or device*, **blocked only** (off). Each entry can be **allowed** or **blocked** directly.
Lists	The block lists in use, each can be switched on/off or removed. Built in: StevenBlack (**on**), Hagezi Light, Hagezi Pro (stricter). Add your own list with *Name* and *List address* (hosts or domain format). **Update now** reloads the lists; they also reload every week.
Own rules	*Always allow* and *Always block*, one domain per line. A blocked domain includes all its subdomains. *Always allow* wins over the lists.
Setup	How to make the devices use the NAS as DNS server, see below.



Setup in short: in the router, enter the address of the NAS as the *local DNS server* (on a FRITZ!Box: Home Network → Network → Network Settings → IPv4 Settings). Then reconnect the devices. The NAS needs a fixed address for this (in the router: "always assign the same IPv4 address", or a static address in section 8). While the NAS is off, devices cannot resolve names until the router falls back to its own DNS, and some routers do not. Single devices can also be switched in their own network settings.

More lists block more, but also more likely something you want. Then add that domain to *Always allow*.

11. Media server and channels

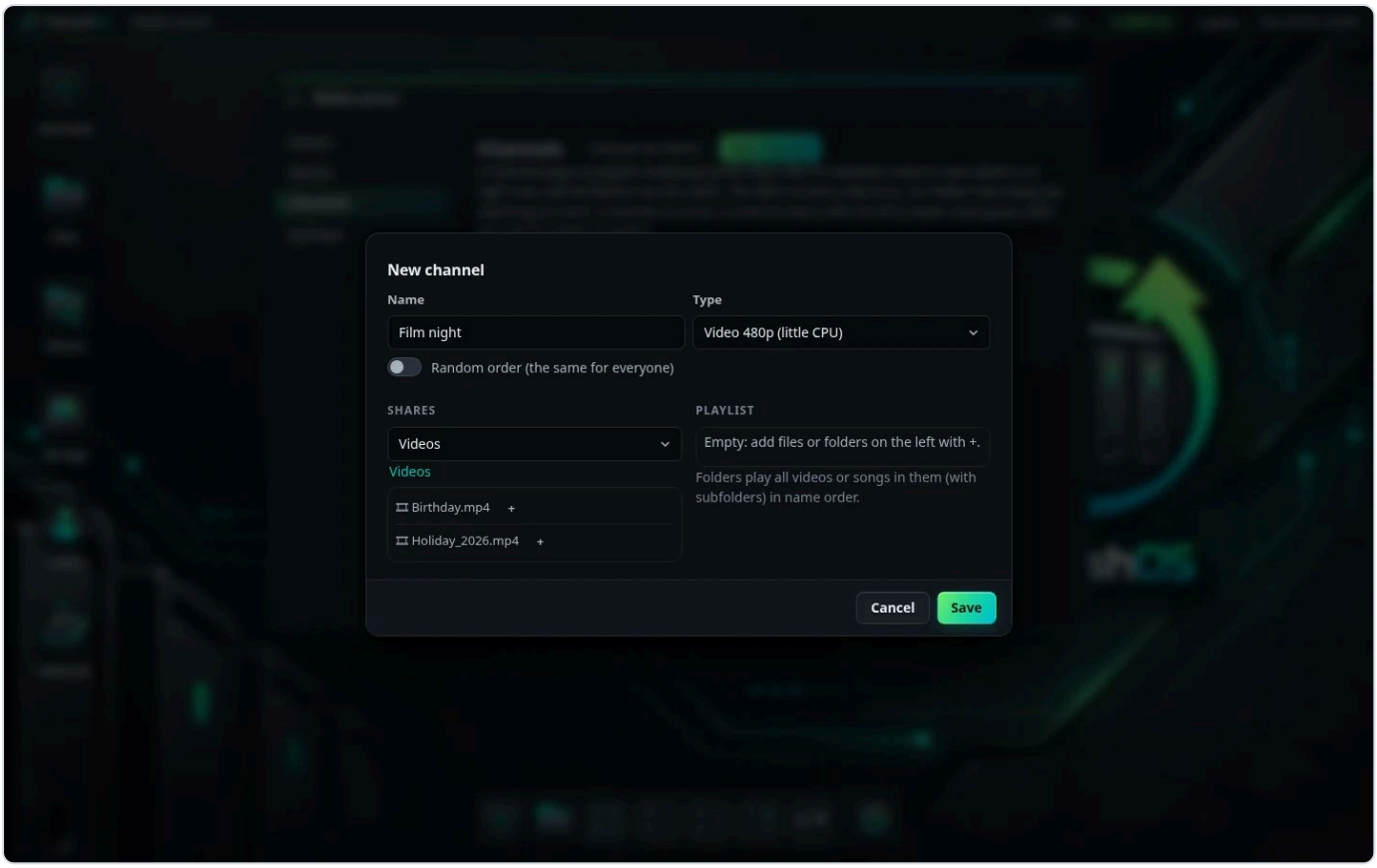
Media server

The media server (DLNA/UPnP) makes videos, music and photos available to TVs, consoles, VLC and Kodi. It appears in the network under the host name of the NAS.

Page	Content and options
Status	State (on/off); number of videos, songs and photos; devices that asked; current playbacks; **Rescan** reads all shares again.
Shares	Per share: Video, Photos, Music. Empty or missing folders are not published. (The same as in Shares.)
Channels	The channels, see below.
Settings	*Media server on* (**off**); *Channels receivable in the home network without signing in* (**on**, for TVs, VLC, Kodi).

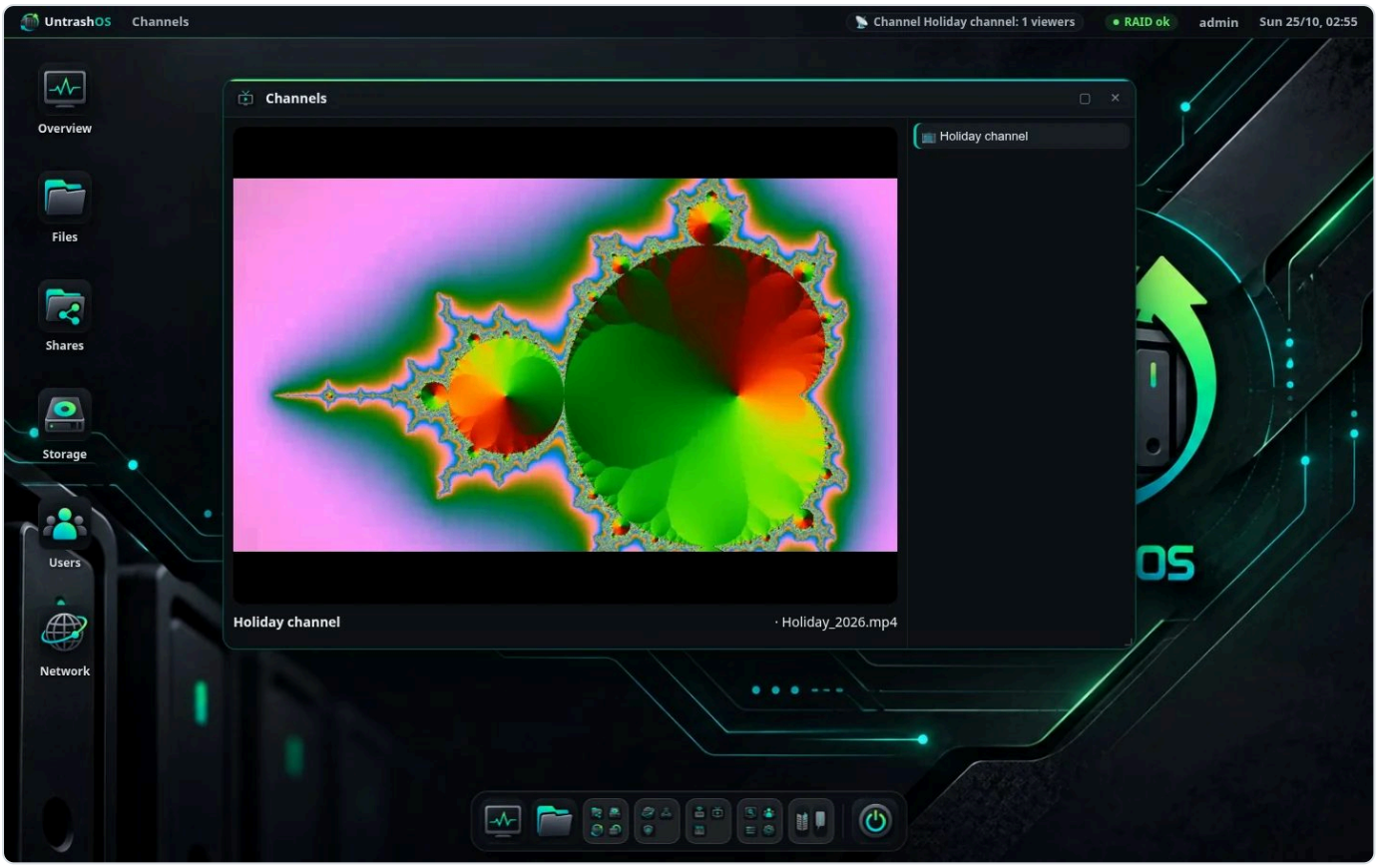
Channels

A channel works like a TV station: it plays its playlist endlessly by the clock, and everyone who tunes in sees the same thing. The NAS converts a channel only once, however many people watch it, and at most 2 channels at the same time. A channel starts with its first viewer and pauses one minute after the last one leaves.

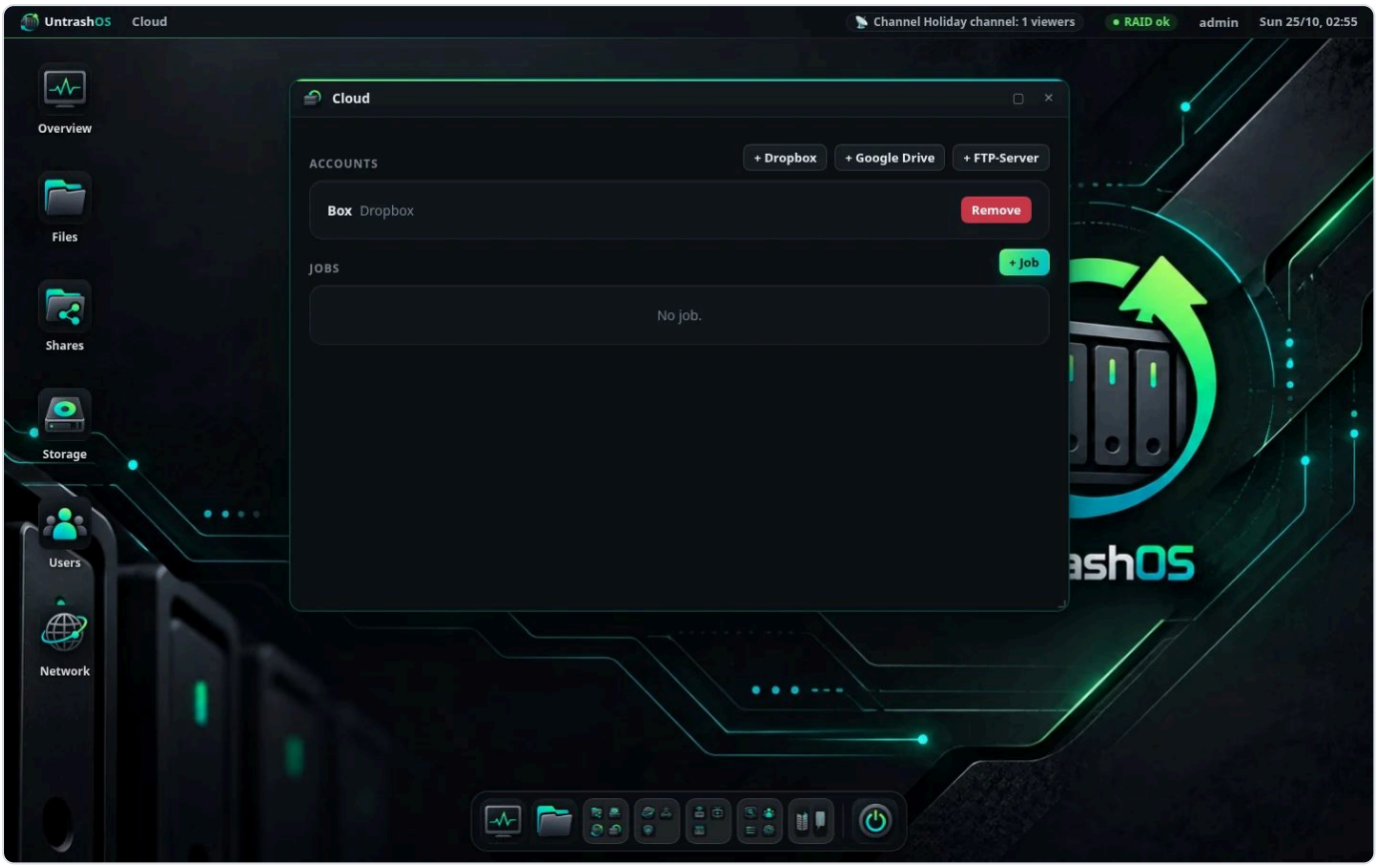


Option	Choices	Meaning
Name	text	Name of the channel.
Type	**Video 480p (little CPU)**, Video 720p (more CPU), Video unchanged (H.264 only, hardly any CPU), Radio (sound only)	Picture quality versus load on the NAS.
Random order	**off**, on	Shuffles the playlist, the same way for every viewer.
Shares / Playlist	files and folders of the shares	Add videos, music or whole folders (**+**). Folders play everything inside them, including subfolders, in name order. Reorder with ↑, remove with ×.

Everyone watches in the **Channels** app (also in the dock of normal users). The list of channels is also available as an M3U playlist for VLC and Kodi (*Channel list (M3U)*).


UntrashOS
Channels
Channel Holiday channel: 1 viewers
RAID ok
admin
Sun 25/10, 02:55
Overview
Files
Shares
Storage
Users
Network
Channels
Holiday channel
Holiday channel
· Holiday_2026.mp4

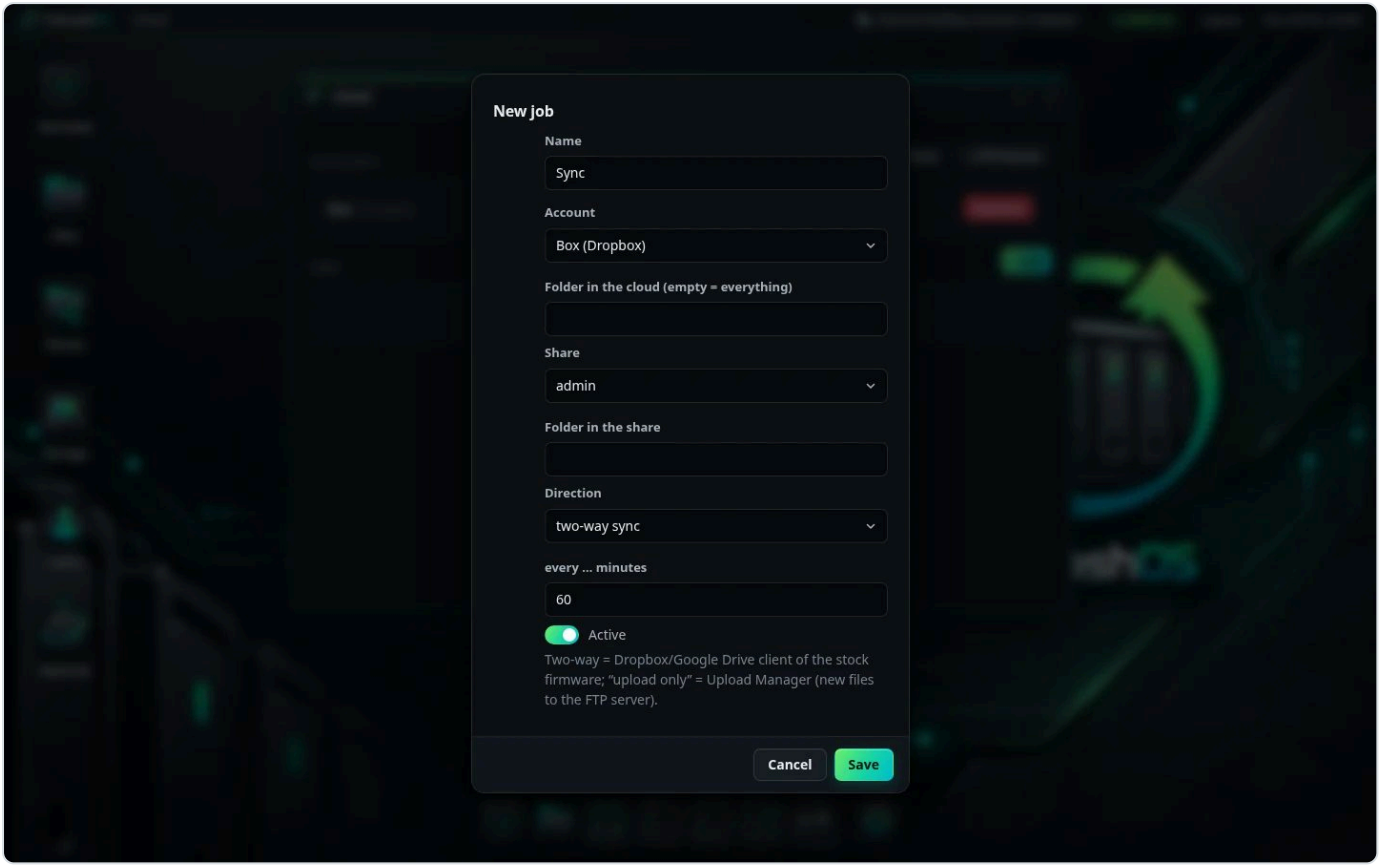
12. Cloud



Accounts

Button	Options	Meaning
+ Dropbox, **+ Google Drive**	Account name	Click **Open the sign-in page**, sign in and agree. The browser then shows an error page under `localhost:53682`; this is expected. Copy the complete address of that page into *Redirect address* and click **Done**.
+ FTP server	Name, Server, Port **21**, User name, Password, TLS (explicit) **off**	An FTP server as target.
Remove		Removes the account. Files stay.

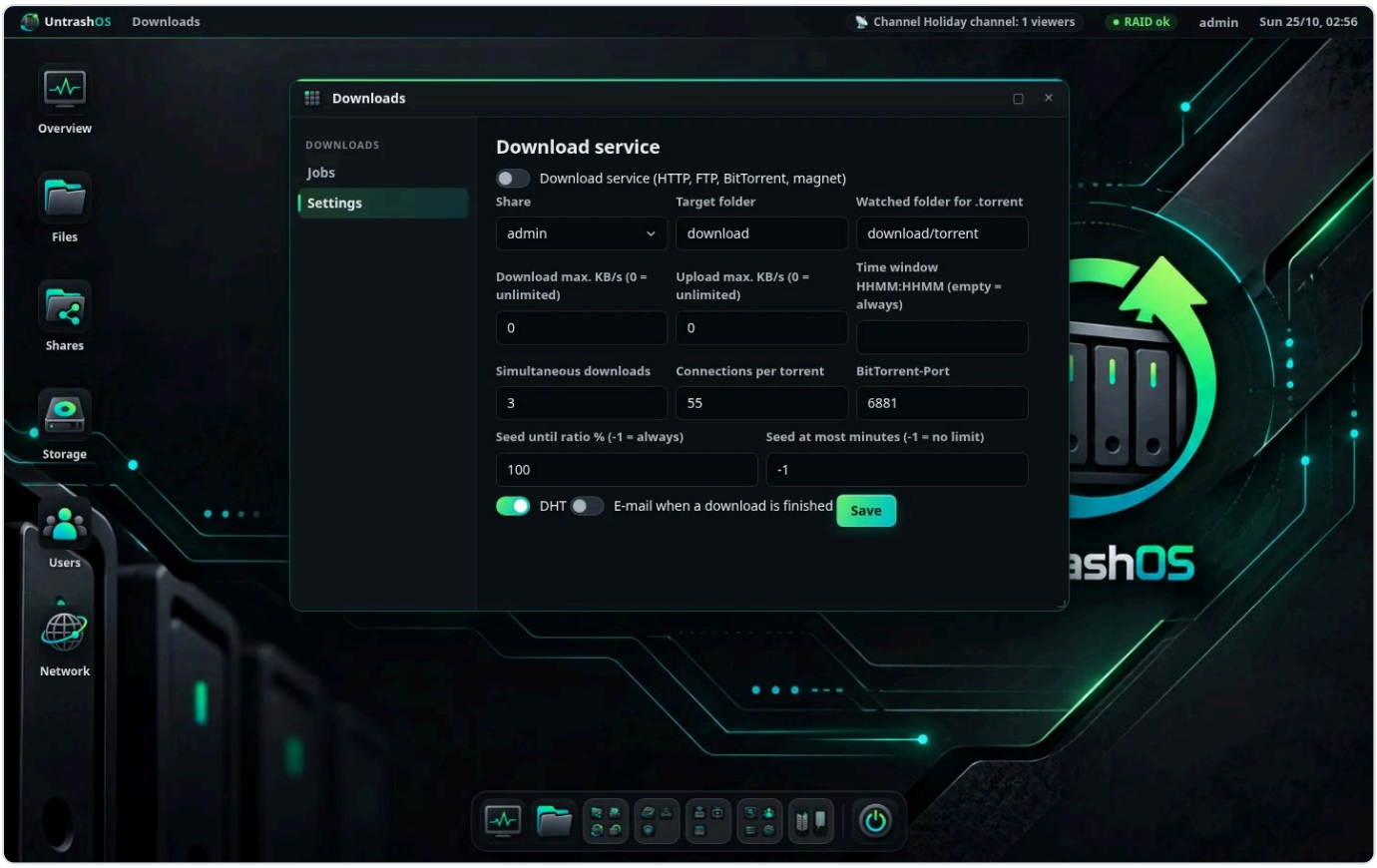
Jobs (+ Job)



Option	Choices	Meaning
Name	default **Sync**	Name of the job.
Account	the connected accounts	
Folder in the cloud	empty = everything	
Share, Folder in the share		The local side.
Direction	**two-way sync**, upload only, download only	*Two-way* works like the Dropbox/Google Drive client of the Zyxel firmware. *Upload only* works like its Upload Manager (new files go to the cloud or FTP server).
every … minutes	default **60**	How often the job runs.
Active	**on**, off	

13. Downloads

The download service loads files over HTTP, FTP, BitTorrent and magnet links into a share.



Jobs lists the downloads with their progress. Add a download by its address (http, ftp or magnet) with **Add**, or upload a `.torrent` file. Downloads can be paused, resumed and removed; *Finished ones from the list* clears the list.

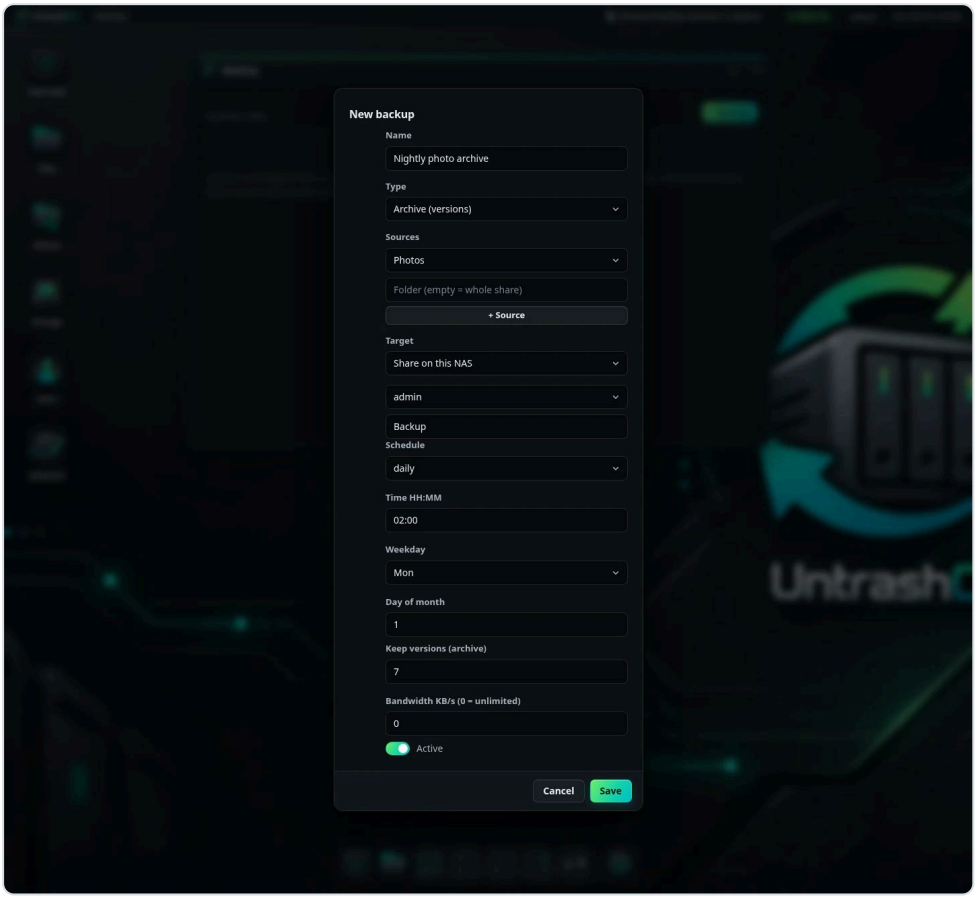
Settings

Option	Choices	Meaning
Download service	**off**, on	Switches the service on.
Share, Target folder	default folder **download**	Where downloads are saved.
Watched folder for .torrent	default **download/torrent**	`.torrent` files put there start automatically.
Download max. KB/s, Upload max. KB/s	**0** = unlimited	Speed limits.
Time window	`HHMM:HHMM`, empty = always	Downloads run only in this time, for example `0100:0700`.
Simultaneous downloads	default **3**	

Option	Choices	Meaning
Connections per torrent	default **55**	
BitTorrent port	default **6881**	
Seed until ratio %	default **100**; -1 = always	Upload share after a torrent is complete.
Seed at most minutes	default **-1** = no limit	
DHT	**on**, off	Finds peers without a tracker.
E-mail when a download is finished	**off**, on	Needs e-mail notification.

14. Backup

+ Backup creates a backup job. The list shows each job with type, target, schedule and last run. Each job can be started at once (**Now**), edited or deleted. **Restore** copies a backup back.



Option	Choices	Meaning
Name	text	
Type	**Archive (versions)**, Synchronization (mirror), Rsync server	*Archive*: one dated folder per run, unchanged files take no extra space. *Synchronization*: the target becomes a mirror of the source. *Rsync*: to an rsync server (another NAS or computer).
Sources	shares, each with an optional folder (empty = whole share); **+ Source** adds more	What is backed up.
Target	**Share on this NAS** (share and folder), connected **USB** drives	Where it goes (archive and synchronization).

Option	Choices	Meaning
Rsync server, Module, User name, Password	only for *Rsync server*	The rsync target. An empty password keeps the saved one.
Schedule	manual only, hourly, **daily**, weekly, monthly; Time **02:00**; Weekday; Day of month	When the job runs.
Keep versions (archive)	default **7**	Older versions are deleted.
Bandwidth KB/s	**0** = unlimited	Limits the speed.
Active	**on**, off	Off keeps the job without running it on schedule.

Restore (archive and synchronization jobs): choose a version, the share to restore into and a folder (default *Restored*). The backup is copied into that folder.

15. Updates



The page shows the installed version and the system slot it runs from (A or B).

Updates from the release page

Once a day the NAS looks on the release page of this project (on github.com) for a newer UntrashOS for this device, downloads it and checks it (signature, checksums). Nothing is installed on its own: when an update is ready, admins see a note after signing in, and the Updates page shows the new version with its release notes.

1. Click **Install UntrashOS x.y** and confirm: the image is written to the **other** slot.
2. Click **Restart now** (or restart later).

Setting	Values	Meaning
Check for updates every day	**on** / off	**Check now** works either way.
Download updates automatically	**on** / off	Off: the page shows the new version with a **Download** button.
Channel	**Stable releases**, Beta (pre-releases too)	Beta also offers test versions before a release.

A check is one anonymous request to github.com; nothing about the NAS is sent. Only images signed by UntrashOS are accepted, and only versions newer than the running one are offered.

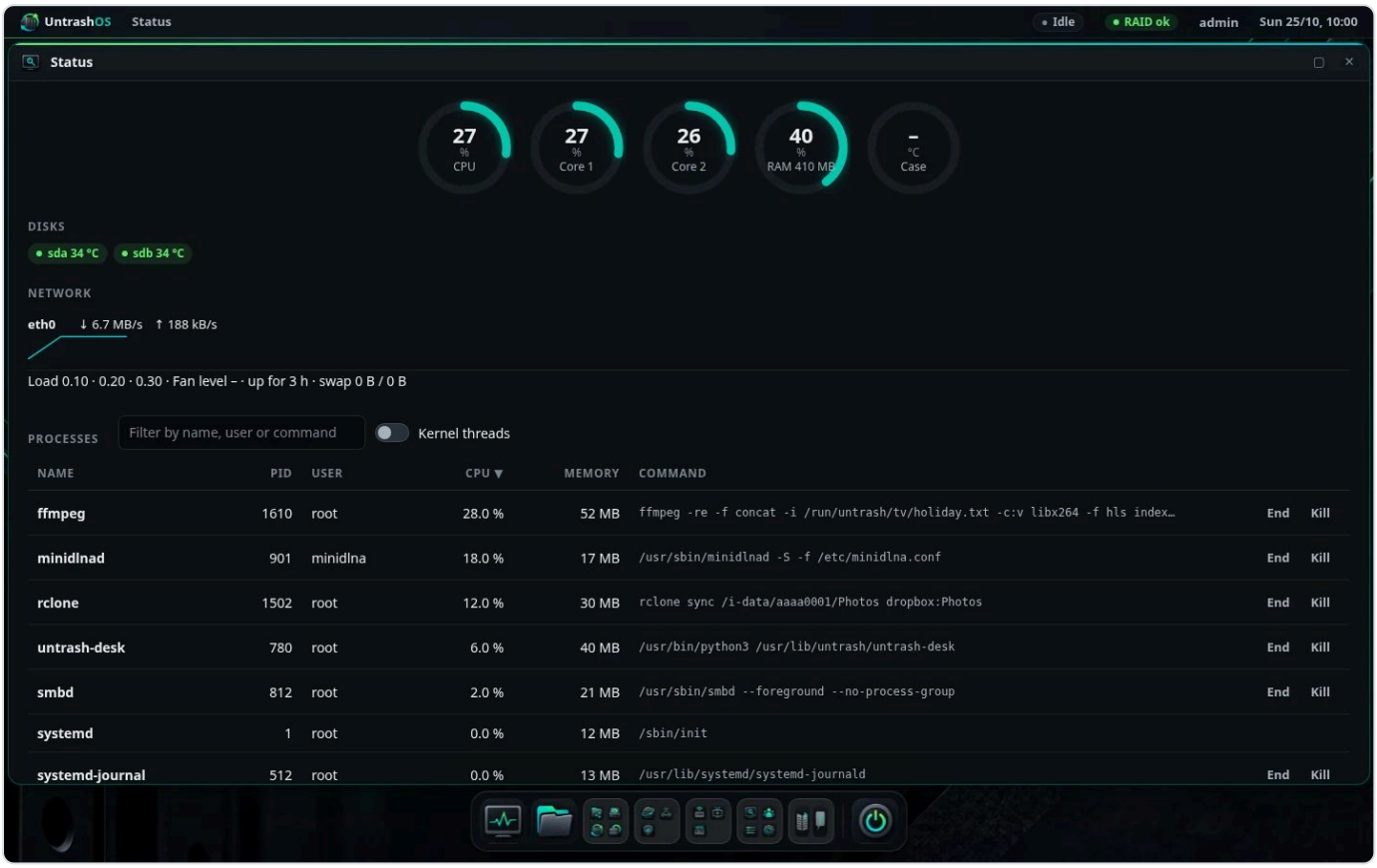
Install from a file

1. Click **Choose a system image …** and select an update file (`untrash-…-update.utslot`). A stick image (`.img`) is refused.
2. The file is uploaded (with progress) and checked: signature and checksums.
3. Click **Install**: the image is written to the **other** slot. An image that is not newer than the running system needs **Install the older version anyway**.
4. Click **Restart now**.

Either way: if the new system does not start cleanly, the loader goes back to the current one by itself. Your settings and data are not part of the slot and stay as they are.

16. Status, logs and system tools

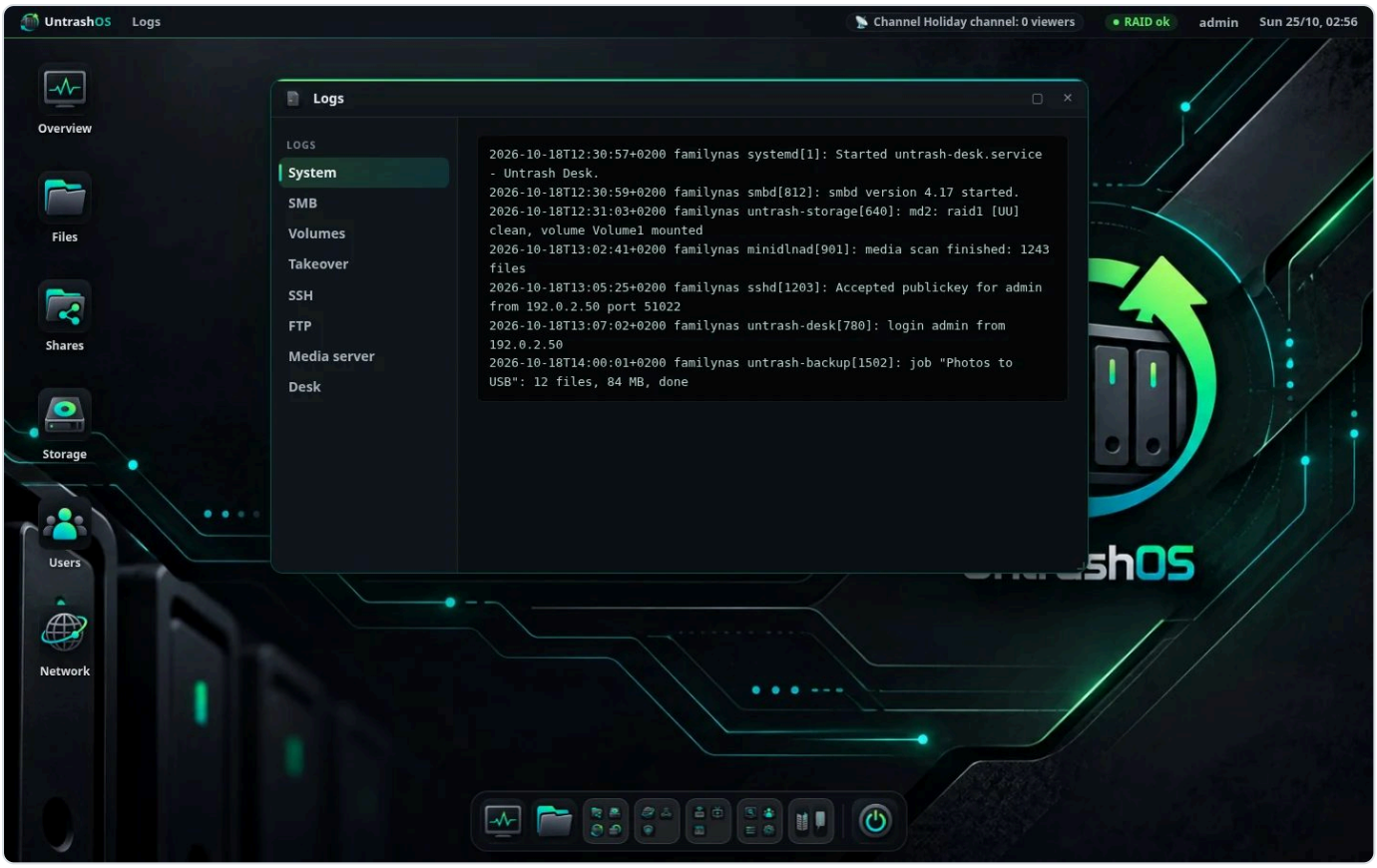
Status is updated every two seconds:



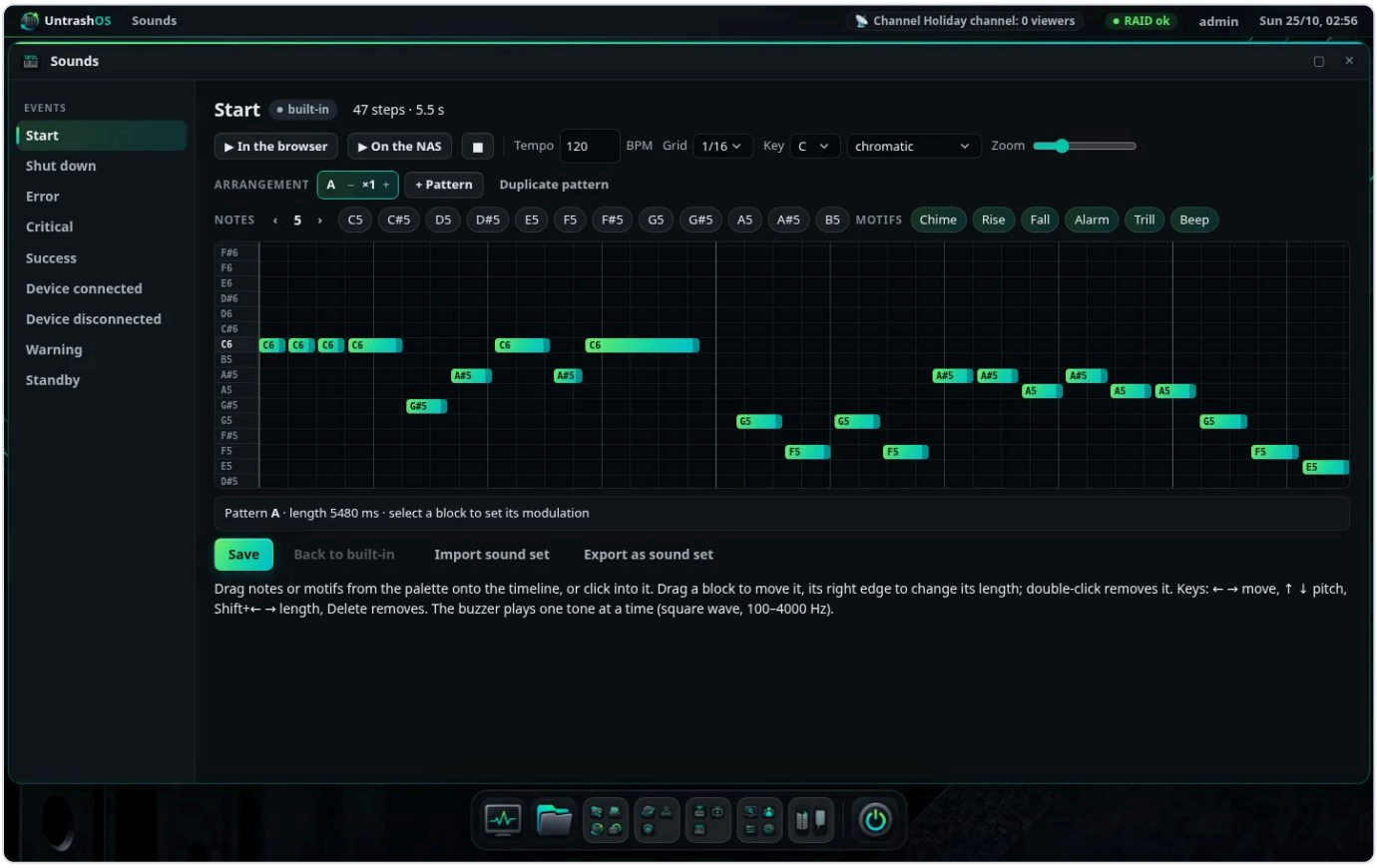
Element	Meaning
Rings	CPU in total and per core, memory, case temperature. Colours change at warning levels.
Disks	Temperature of each disk. Sleeping disks are shown as asleep and are not woken up.
Network	Current traffic per port.
Bottom line	Load, fan level, uptime, swap in use.
Processes	Every running program with its user, CPU share, memory and command line. Click a column header to sort, the filter field searches name, user and command; **Kernel threads** also shows the kernel's own threads.

End asks a program to stop (it can still save and clean up); **Kill** stops it at once, without saving anything. Both ask first. Services of the system are usually started again automatically. PID 1, kernel threads and the Desk itself cannot be ended here.

Logs shows the system log or the log of a single service: System, SMB, Volumes, Takeover (the first start after way B), SSH, FTP, Media server, Desk.



Sounds edits the melodies the NAS buzzer plays for events: Start, Shut down, Error, Critical, Success, Device connected, Device disconnected, Warning, Standby. Events you changed are marked ★.



The editor works like a small loop-based music program. The buzzer plays one tone at a time (square wave, 100–4000 Hz), so blocks on the timeline never overlap.

Part	Meaning
▶ In the browser, ▶ On the NAS, ■	Plays the whole melody, or stops it. The browser plays exactly what the buzzer plays.
Tempo	Beats per minute (40–300). Changing it makes the melody faster or slower; its shape stays.
Grid	Note lengths snap to 1/8, 1/16 or 1/32 of a bar.
Key, scale	The rows of the timeline: only the notes of this key and scale (major, minor, pentatonic, minor pentatonic, blues, chromatic). Older melodies open in the key that holds all their notes.
Arrangement	The order of the patterns, each with its number of repeats (**−**, **+**). ◀ moves a pattern earlier, ✕ takes it out. **+ Pattern** adds an empty one, **Duplicate pattern** a copy of the pattern you edit. Click a pattern's name to edit it.
Notes, Motifs	Drag a note (choose the octave with **‹ ›**) or a motif (Chime, Rise, Fall, Alarm, Trill, Beep) onto the timeline.
Timeline	Click an empty place for a new note. Drag a block to move it in time and pitch, drag its right edge to change its length, double-click to remove it. The orange line is the end of the pattern; drag it to leave a rest before the next repeat. Keys: **←** **→** move, **↑** **↓** pitch, **Shift** + **←** **→** length, **Delete** removes, **Ctrl** + **D** duplicates.
Block below the timeline	Note, length and **modulation** of the selected block: *Staccato* (short, with a gap), *Vibrato* (depth, rate), *Slide* (glide by up to two octaves), *Fall* (drops an octave at the end), *Arpeggio* (major, minor, fifth or octave chord, speed).
Save, Back to built-in	Saves your melody (at most 60 s), or returns to the original.
Import sound set, Export as sound set	Loads or saves all melodies as a JSON file; the editor's patterns are part of it.

Terminal (administrators only) opens a shell on the NAS in the browser window.

UntrashMesh connects several UntrashOS devices to one encrypted network (WireGuard), at home and over the internet. No cloud provider and no fixed IP address are needed: every device keeps an address book of the others, and a device whose address changes tells all members at once; members also pass such news on, so it reaches devices the changed one cannot reach itself.

Action	What happens
Create the mesh	This NAS starts a mesh and is its admin.

Action	What happens
Invite a device	Shows an invitation text (valid once, 24 hours). Give it to the other device.
Join a mesh	Paste an invitation. The device becomes a member.
Share… (per member)	Offers shares of this NAS to that member, **read only** or **read/write**.
Remove (admins)	Takes a member out; its connections and offered shares end.
Leave the mesh	This NAS leaves; to come back it needs a new invitation.

Members reach each other by name: `\\<name>.mesh` for shares, `http://<name>.mesh` for the Desk. Over the internet at least one side needs a router with UPnP, or a forwarding of UDP port 51820; UntrashOS asks the router itself when UPnP is on.

Partner volumes (co-op mode). Shares a member offers appear on this NAS as partner volumes: under **Storage** and in the **UntrashMesh** window. Under **Shares** you can make shares on them for your own users, like on any volume. The member decides what you may do (read only or read/write); partner volumes are reachable only over UntrashMesh, and they are unavailable while the member is offline. Administrators also find them in **Files** under **Partner volumes** and can work on them there directly.

Off-site mirror. A member can keep a complete copy of a data volume of this NAS (RAID 1 over UntrashMesh), so the data survives a fire or a theft at one place. In the **UntrashMesh** window choose **Mirror a volume…**, the volume and the member. The member's admin must agree and choose a volume for the copy, with as much free space as your volume is large. The mirror starts with the next restart of this NAS.

- Writes go to your volume first and to the partner as fast as the line allows. While the partner is not reachable, the changes are kept and sent when it is back.
- Your volume keeps its format: without the mirror it is mounted as before.
- If the mirror cannot start at a restart, the volume is mounted directly and the copy is sent again completely once the partner is reachable (**needs a full resync**).
- The volume of the running system cannot be mirrored.
- The copy is not encrypted on the partner (the connection is): choose a member whose administrator you trust with the data.
- **End** (either side): your volume stays as it is and is mounted without the mirror from the next restart on; the partner deletes its copy.

Office. Documents stay on the NAS and are edited in the browser by an office server that runs elsewhere (Collabora Online, Nextcloud Office or ONLYOFFICE; it needs more memory than the NAS has). Under **Settings → Office** an admin enters the address of the office server and the address under which that server reaches this NAS (for example `http://untrash.local`, or `http://<name>.mesh` over UntrashMesh). **Save and check** lists the document types the office server supports. Files then shows **Open in Office** for them; the document opens in a new tab and is saved back to the NAS. In the editor, **Save As** and **Export As** (for example as PDF) ask for a name and put the copy in the same folder on the NAS. Everyone keeps the rights of the share: with read-only access the document opens read-only. The office server must allow this NAS as storage; see its documentation. Without this setting, a Nextcloud can also mount the shares as External storage (WebDAV, see Services, or SMB).

Nextcloud. A Nextcloud with the UntrashOS app (in development, not released yet) can join UntrashMesh as a member. An admin chooses **UntrashMesh → Connect a Nextcloud**, picks the shares the Nextcloud may follow and gives the invitation (valid once, 24 hours) to the Nextcloud admin, who pastes it under **Administration settings → UntrashOS**. While it joins, the Nextcloud must reach this NAS (same network, or over the internet with UPnP on the router).

- The Nextcloud keeps the mesh's address book at its internet address. When members lose each other, for example because the internet address at home changed, they find each other again through it. It only stores what the members signed, so it cannot redirect anyone; it does see their addresses, so connect a Nextcloud you trust.
- This NAS tells the Nextcloud which files changed in the shares it follows. The Nextcloud then does not have to scan the shares, and the disks can sleep.
- The Nextcloud reads the files over WebDAV (switch it on under **Settings → Services**), with the rights of each user.
- **Shares…** (per Nextcloud) changes the shares it follows on this NAS. On other NAS of the mesh an admin chooses them there.

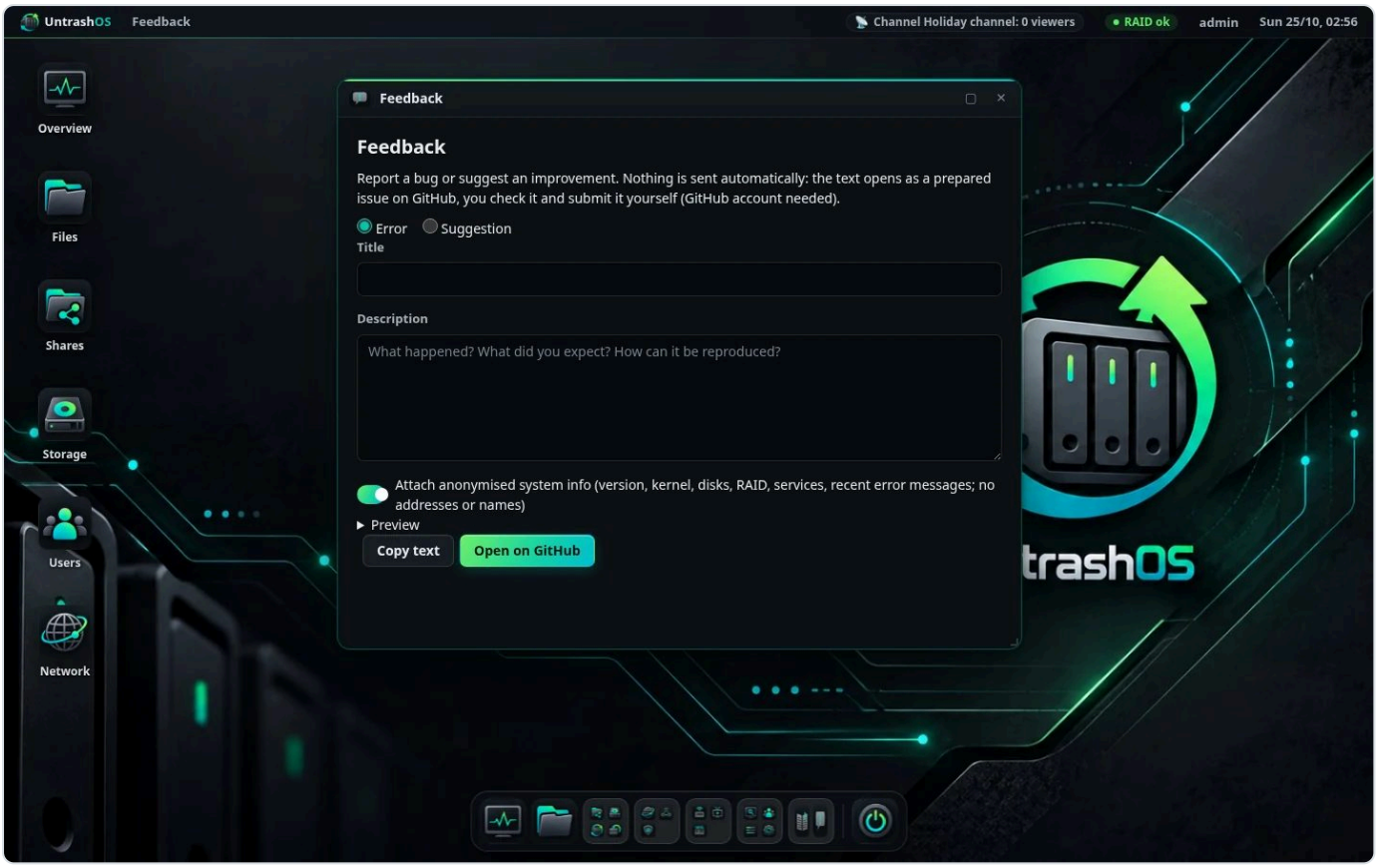
UntrashApp. The UntrashApp for phones (in development, not released yet) joins UntrashMesh and signs in on the NAS as one Desk user. An admin pairs it under **UntrashMesh → Pair the UntrashApp**: choose the user, then scan the QR code with the app (valid once, for one hour). On other NAS of the mesh an admin links the app to a user there with **Link…**.

Under **Account**, every user with a paired app chooses how to sign in (the password is needed to change it):

Sign in with	What happens
Password only	as before
Password or the UntrashApp	on the sign-in page, **Sign in with the UntrashApp** shows a number; choose the same number in the app
Password and the UntrashApp	after the password the app must confirm as well; installing updates, deleting users and similar actions also need a confirmation in the app

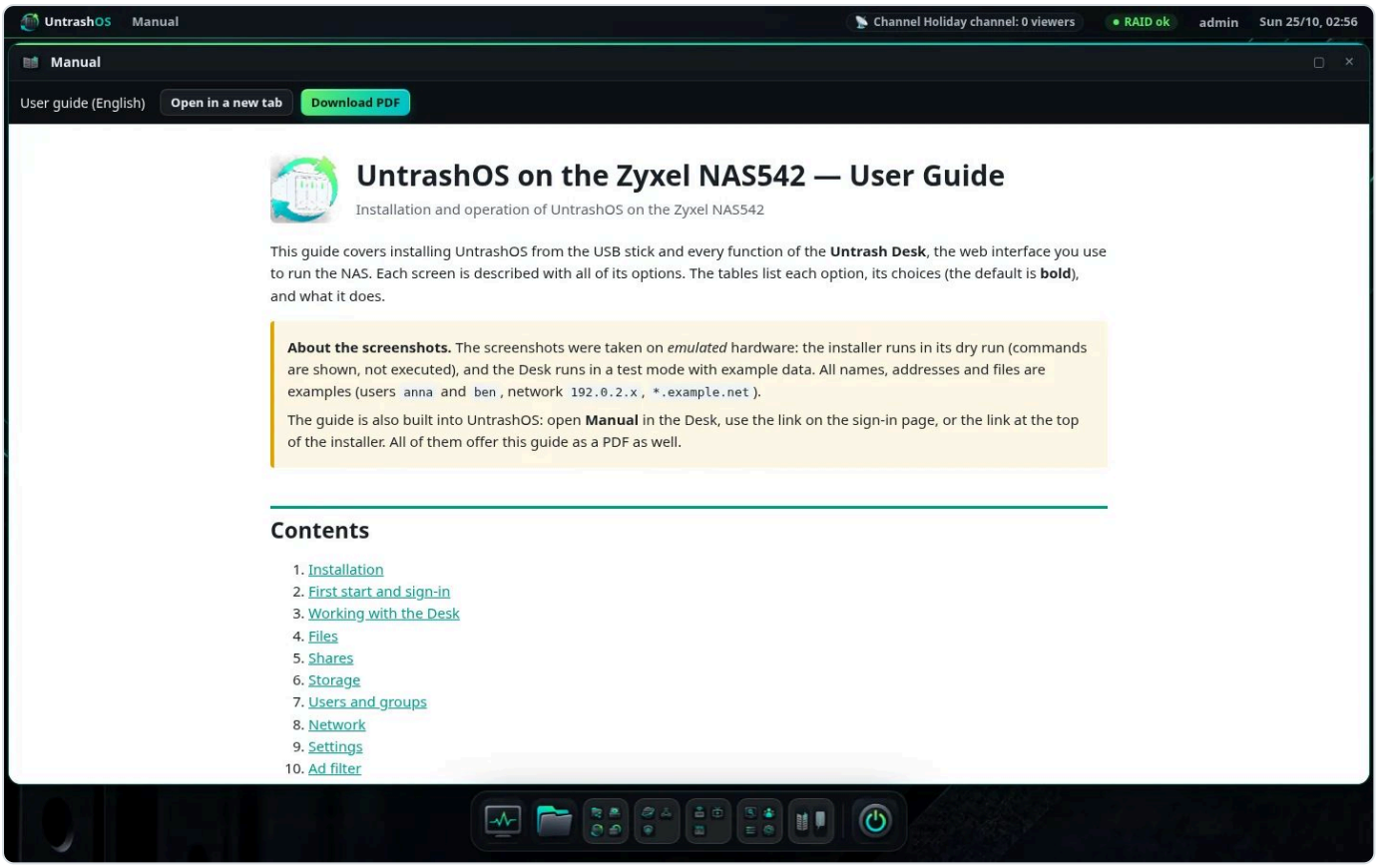
Lost phone: an administrator chooses **Reset to password** for that user under **Users**. For the admin account, the RESET button on the NAS (admin password reset) also resets the sign-in.

Feedback prepares a bug report or a suggestion as a GitHub issue. Nothing is sent automatically.

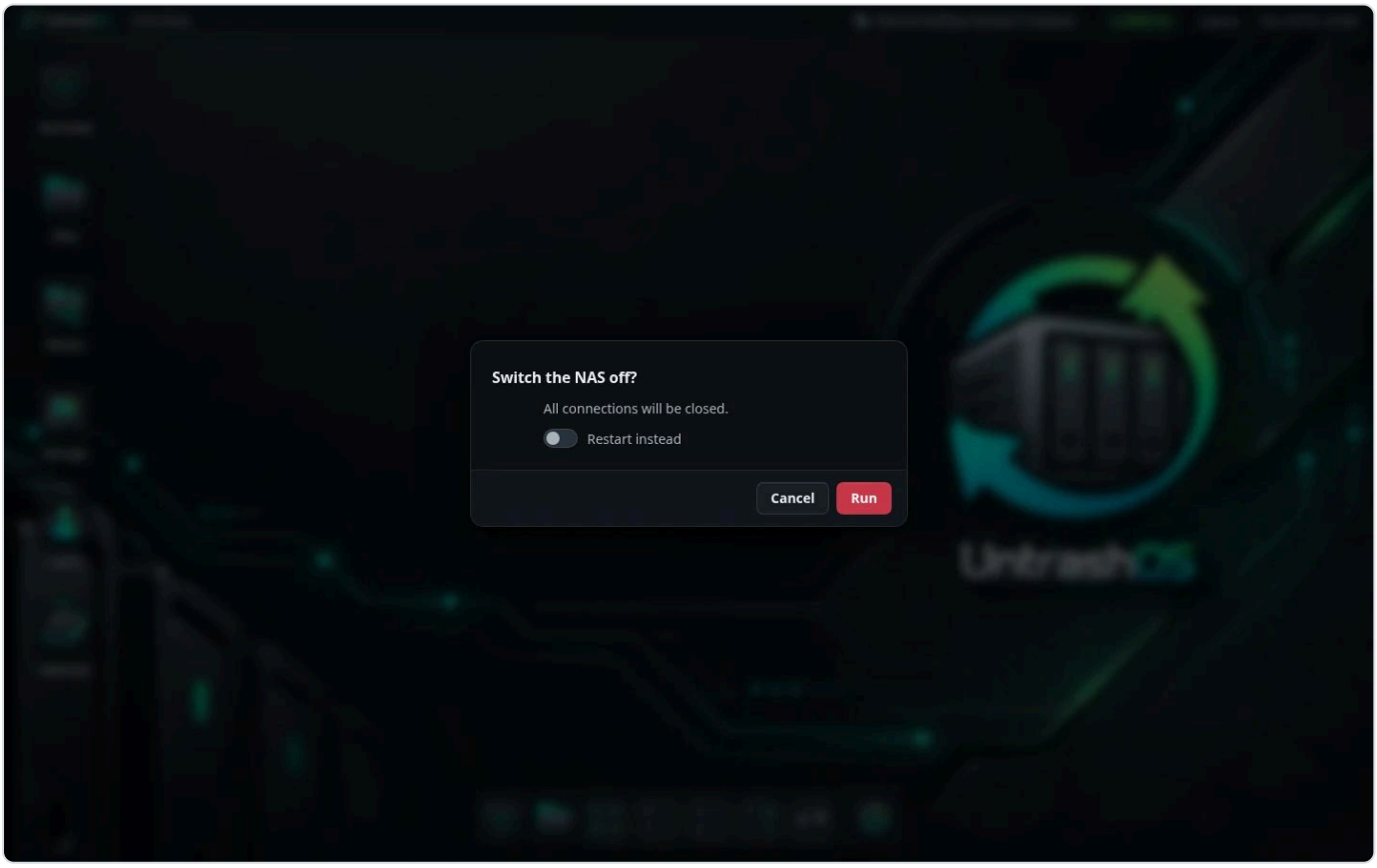


Option	Choices	Meaning
Kind	**Error**, Suggestion	
Title, Description	text	What happened, what you expected, how to reproduce it.
Attach anonymised system info	**on**, off	Version, kernel, disks, RAID, services and recent error messages, without addresses or names. **Preview** shows exactly what is attached.
Copy text, Open on GitHub		Copies the text, or opens it as a prepared issue on GitHub. You check it and submit it yourself (GitHub account needed).

Manual shows this guide inside the Desk, with **Open in a new tab** and **Download PDF**:



Power button (end of the dock): **Run** switches the NAS off. With *Restart instead* it restarts. All connections are closed.



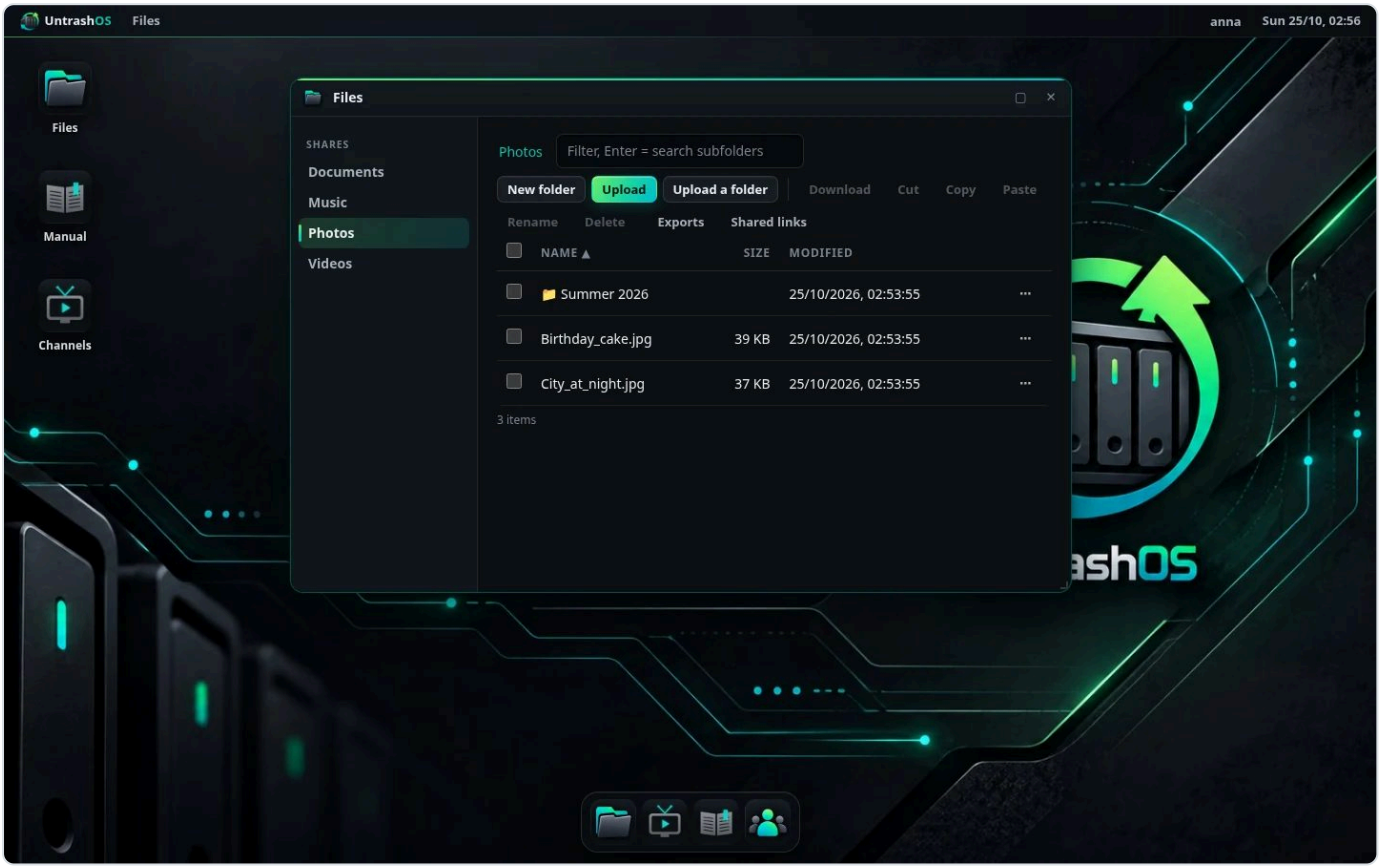
Your user name (top bar) opens **Account**: change the *language of this browser* (— as system —, English, Deutsch, Français), the **Appearance**, or **Sign out**.

Appearance belongs to your user: it is stored on the NAS and follows you to every browser you sign in with. It applies at once.

Setting	Values	Meaning
Theme	Automatic (as the system), **Dark**, Light, High contrast	*Automatic* follows the light/dark and "more contrast" setting of your computer or phone. *High contrast*: black background, white text, strong borders, no transparency or wallpaper.
Colour vision	**Normal colour vision**, Red-weak (protanopia), Green-weak (deuteranopia), Blue-weak (tritanopia), No colour vision (achromatopsia)	Other colours for states (healthy, warning, error, information) that stay apart for that kind of colour vision, plus a symbol on each state (✓ ! ✕ i) and stripes on warning and error bars, so that nothing depends on colour alone.
Reduce motion	**off** / on	No animations (also on when your system asks for reduced motion).

17. Working as a normal user

Users who are not administrators see a smaller Desk: **Files**, **Channels**, **Manual** and **My account**. They see only the shares they have rights to and can only change what those rights allow.



My account: *Current password*, *New password*, *Repeat*, then **Change password**. **Sign out** ends the session. **Appearance** below works as described in section 3.

18. Back to the Zyxel firmware

UntrashOS can always go back to the original Zyxel firmware with the backup on the installation stick. Data on the disks stays. The UntrashOS files in `/.untrash` (ways B and C) or the UntrashOS disk (way A) are left in place, so you can return to UntrashOS later.

From the Desk: plug in the installation stick, open **Settings → Back to stock**. The page shows the backup found on the stick. Type `STOCK` and click **Back to the Zyxel firmware**.

From the installer (if UntrashOS does not start any more): start the NAS with the installation stick and choose way D.

Both ways then do the same:

1. The backup is checked against its checksums, and the stick gets the `ROLLBACK` order.
2. The NAS restarts and runs the Zyxel firmware once. The firmware writes its boot data back from the backup and switches the NAS off after a few minutes. Leave the stick in.
3. Switch the NAS on: it starts the Zyxel firmware as before the installation.